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Chapter 1

Metric Spaces

1.1 Topological Spaces

Definition 1.1. Given a set X, a topology τ is a subset of the power set $\mathcal{P}(X)$ such that:

i) $\emptyset, X \in \tau$.

ii) τ is closed under union, i.e., if $E_\alpha \in \tau$ for $\alpha \in I$ then $\bigcup_{\alpha \in I} E_\alpha \in \tau$.

iii) τ is closed under finite intersection, i.e., $E_1, E_2, \ldots E_n \in \tau \implies \bigcap_{i=1}^{n} E_i \in \tau$.

Example 1.1. Some examples of topologies are as follows:

- $\tau_{disc} = \mathcal{P}(X)$.
- $\tau_{trivial} = \{\emptyset, X\}$.

Definition 1.2. Let (X, τ) be a topological space with $A \subseteq X$. We have:

1. A is open if and only if $A \in \tau$.
2. A is closed if and only if $A^c \in \tau$.

Definition 1.3. Let (X, τ) be a topological space. A neighborhood of a point $x \in X$ is an open set containing x.

Definition 1.4. Let (X, τ) be a topological space and $A \subseteq X$. We say that x is an interior point of A if and only if there exists an open neighborhood U_x of x contained in A. We denote the set of interior points of A by A°.

Definition 1.5. Let (X, τ) be a topological space. We say that a point $x \in X$ is a limit point of A if and only if for every open neighborhood U_x of x we have

$$U_x \setminus \{x\} \cap A \neq \emptyset.$$

We denote the set of limit points of A by A'.

Definition 1.6. Let (X, τ) be a topological space with $A \subseteq X$. The closure of A is defined as the set $\overline{A} = A \cup A'$.

We have the following properties.

Proposition 1.1. *1. A° is open; in fact, it is the largest open set contained in A, so A is open if and only if $A^\circ = A$.*

2. $\overline{A}$ is closed; in fact, it is the smallest closed set containing A, so A is closed if and only if $A = \overline{A}$ if and only if $A' \subseteq A$.

Proof of 1. If $x \in A^\circ$, then there exists $U_x \subseteq A$. It is clear that $A^\circ \subseteq \bigcup_{x\in A^\circ} U_x$. For the other inclusion, notice that if $y \in U_x$, then U_x is a neighborhood of y and is contained in A, implying that $y \in A^\circ$ and so $U_x \subseteq A^\circ$ for every x. We deduce that $A^\circ = \cup_{x\in A^\circ} U_x$ and so A° is open, being a union of open sets.

Assume G is an open set contained in A. For $x \in G$, there exists U_x such that $U_x \subseteq G \subseteq A^\circ$, and so $x \in A^\circ$ which gives $G \subseteq A^\circ$. Thus, A° is the largest open set contained in A.

Finally, by definition, we have $A^\circ \subseteq A$. If A is open, then by the above, $A \subseteq A^\circ$. Hence $A = A^\circ$.

The proof of 2 is left as an exercise. □

Remark 1.1. As a consequence of the above, A is open if and only if for every $x \in A$ there exists a neighborhood U_x of x such that $U_x \subseteq A$.

1.1.1 Continuity

Definition 1.7. Let (X, τ_X) and (Y, τ_Y) be two topological spaces, and let $f : X \to Y$ be a function, and let $x_0 \in X$. f is continuous at x_0 if and only if for every open neighborhood V of $f(x_0)$ in Y there exists an open neighborhood U of x_0 in X such that $f(U) \subseteq V$.

We say that f is continuous on X if it is continuous at every point of X.

Proposition 1.2. *Let $f : X \to Y$ be a function between two topological spaces. Then f is continuous if and only if $f^{-1}(G)$ is open in X for every open set G in Y.*

1.1.2 Sequences

Definition 1.8. Let (X, τ) be a topological space, and let x_n be a sequence in X. We say that x_n converges to $x_0 \in X$ if and only if for every neighborhood U_{x_0} of x_0 there exists $N \in \mathbb{N}$ such that

$$x_n \in U_{x_0} \quad \forall n \geq N.$$

Definition 1.9. Take the sequence $x_n = \frac{1}{n}$.

- In $(\mathbb{R}, \tau_{disc})$ it does not converge.
- In $(\mathbb{R}, \tau_{trivial})$ it converges to every point in $\mathbb{R}$.

Proposition 1.3. *Let (X, τ) be a topological space, and let x_n be a sequence in A converging to x_0. Then $x_0 \in \overline{A}$.*

Proof. Suppose that $x_0 \notin \overline{A}$. Then there exists a neighborhood U_{x_0} of x_0 such that $U_{x_0} \subseteq (\overline{A})^c = A^c \cap (A')^c$. Since x_n is a sequence in A converging to x_0, $x_n \in U_{x_0}$ for $n \geq N$ for some N, a contradiction since $x_n \in A$. □

Proposition 1.4. *Let $f : (X, \tau_X) \to (Y, \tau_Y)$ be a function between two topological spaces, and let $x_0 \in X$, with f continuous at x_0. If x_n converges to x_0, then $f(x_n)$ converges to $f(x_0)$.*

Remark 1.2. The converse of both propositions above is not true in general.

1.1.3 Separability

Definition 1.10. Let (X, τ) be a topological space and let $A \subseteq X$. We say that A is dense in X if and only if $\overline{A} = X$.

Definition 1.11. Let (X, τ) be a topological space. We say that X is separable if and only if X has a countable dense subset.

Example 1.2. Some familiar examples are:

- $(\mathbb{R}, |\cdot|)$ is separable since $\mathbb{Q}$ is dense in $\mathbb{R}$.
- If X is uncountable, (X, τ_{disc}) is not separable since $\overline{A} = A$ for every $A \subseteq X$ so the only dense subset of X is X itself.

1.1.4 Hausdorff topology

Definition 1.12. Let (X, τ) be a topological space. We say that X is Hausdorff if and only if for every pair of distinct points $x, y \in X$, there exist an open neighborhood U_x of x and an open neighborhood U_y of y such that

$$U_x \cap U_y = \emptyset$$

.

Proposition 1.5. *The limit of a convergent sequence in a Hausdorff space is unique.*

Proof. Assume that (x_n) converges to x and y with $x \neq y$. Since (X, τ) is Hausdorff, there exist neighborhoods U_x of x and U_y of y such that

$$U_x \cap U_y = \emptyset.$$

Since (x_n) converges to x, there exists $N_1 \in \mathbb{N}$ such that $x_n \in U_x$ for all $n \geq N_1$. In particular, this implies that $x_n \notin U_y$ for all $n \geq N_1$, contradicting the fact that (x_n) also converges to y. □

1.1.5 Metric Topology

Definition 1.13. Given a set X, a metric d on X is a function

$$d : X \times X \to \mathbb{R}$$

satisfying the following

i) $d(x, y) \geq 0$ for all $x, y \in X$ with equality if and only if $x = y$.

ii) $d(x, y) = d(y, x)$ for all $x, y \in X$.

iii) $d(x,z) \leq d(x,y) + d(y,z)$ for all $x, y, z \in X$.

We say that (X,d) is a metric space.

Definition 1.14. Given a metric space (X,d), for $x \in X$ and $r > 0$, we define the ball of radius r centered at x as follows

$$B(x,r) = \{y \in X : d(x,y) < r\}.$$

We define $\tau_d = \{U \in \tau_d : \forall x \in U \, \exists r > 0 \, s.t. \, B(x,r) \subseteq U\}$. It is easy to check that τ_d is a Hausdorff topology on X; in fact, it is the smallest topology that contains all balls.

We say that τ_d is the metric topology induced by d.

Definition 1.15. Given a metric space (X,d), we say that $E \subseteq X$ is bounded if and only if $E \subseteq B(x_0, R)$ for some $x_0 \in X$ and $R > 0$. In particular, a sequence is bounded if and only if there exists $x_0 \in X$ and $R > 0$ such that $d(x_n, x_0) < R$ for all $n \in \mathbb{N}$.

Proposition 1.6. *Let (X,d) be a metric space. A sequence x_n converges to x_0 if and only if $\forall \varepsilon > 0 \quad \exists N \in \mathbb{N}$ such that $d(x_n, x_0) < \varepsilon \quad \forall n \geq N$.*

Proposition 1.7. *Let (X, d_X) and (Y, d_Y) be two metric spaces, and let $f : X \to Y$ with $x_0 \in X$. f is continuous at x_0 if and only if $\forall \varepsilon > 0 \quad \exists \delta > 0$ such that $d_Y(f(x), f(x_0)) < \varepsilon$ whenever $d_X(x, x_0) < \delta$ if and only if for every sequence x_n converging to x_0 we have $f(x_n)$ converges to $f(x_0)$.*

Proposition 1.8. *Let (X,d) be a metric space and $E \subseteq X$. Then $x \in \overline{E}$ if and only if there exists a sequence $x_n \in E$ such that x_n converges to x.*

Proof. ($\Longleftarrow$) This is true for any topological space.
($\Longrightarrow$) Let $x \in \overline{E} = E \cup E'$. Then either $x \in E$ or $x \in E'$. If $x \in E$, we may take the constant sequence $x_n = x$ for all n. If $x \in E'$, then by definition of a limit point, for every $n \in \mathbb{N}$ there exists $x_n \in B(x, \frac{1}{n}) \setminus \{x\} \cap E$. Hence $d(x_n, x) < \frac{1}{n}$, so $x_n \to x$ by the Archimedean property. □

We then have the following corollary.

Corollary 1.1. *Given a metric space (X,d) with $E \subseteq X$, E is closed if and only if for every convergent sequence x_n with $x_n \in E$ we have $\lim_{\to\infty} x_n \in E$*

Example 1.3. Some examples of separable and nonseparable metric spaces are:

- $(\mathbb{R}^n, |\cdot|)$ is separable since $\mathbb{Q}^n$ is dense in $\mathbb{R}^n$.
- The space $C([a,b])$ of real-valued continuous functions on $[a,b]$ with the sup-metric (uniform convergence) $d_\infty(f,g) = \sup_{x \in [a,b]} |f(x) - g(x)|$. Let $\mathcal{P}(\mathbb{R}) = \{a_k x^k + \cdots + a_1 x + a_0 : a_i \in \mathbb{R}\}$ be the space of real polynomials. By the Weierstrass approximation theorem, for every $f \in C([a,b])$ there exists a sequence of polynomial functions p_n such that $p_n \to f$ uniformly on $[a,b]$, i.e.

  $$\lim_{n \to \infty} d_\infty(p_n, f) = 0.$$

  For $\varepsilon > 0$, choose $p \in \mathcal{P}(\mathbb{R})$ such that

  $$d_\infty(p, f) < \frac{\varepsilon}{2}.$$

Write

$$p(x) = c_k x^k + \cdots + c_1 x + c_0,$$

with $c_i \in \mathbb{R}$. For each coefficient c_i, choose $q_i \in \mathbb{Q}$ such that

$$|c_i - q_i| < \frac{\varepsilon}{2(k+1)M}, \qquad M = \max_{x\in[a,b]} \{1, |x|, \dots, |x|^k\}.$$

Define

$$q(x) = q_k x^k + \cdots + q_1 x + q_0.$$

Then

$$\begin{aligned} |p(x) - q(x)| &\le \sum_{i=0}^{k} |c_i - q_i|\, |x|^i \\ &\le \sum_{i=0}^{k} \frac{\varepsilon}{2(k+1)M}\, |x|^i \\ &\le \frac{\varepsilon}{2(k+1)M}\, (k+1)M \\ &= \frac{\varepsilon}{2}. \end{aligned}$$

Hence $d_\infty(q, f) \le d_\infty(q, p) + d_\infty(p, f) < \varepsilon$.

We conclude that the countable set $\mathcal{P}(\mathbb{Q})$ is dense in $C([a, b])$ so $C([a, b])$ is separable.

- We define the space of bounded sequences $\ell^\infty(\mathbb{R}) = \{(x_1, x_2, \dots) : x_n \in \mathbb{R},\ \sup_n |x_n| < \infty\}$, with the metric

$$d_\infty(x, y) = \sup_{i\in\mathbb{N}} |x_i - y_i|.$$

Then $\ell^\infty(\mathbb{R})$ is not separable. Suppose, for contradiction, that it is separable, and let M be a countable dense subset. Consider

$$A = \{0, 1\}^{\mathbb{N}},$$

the set of all sequences with entries 0 or 1. The set A is uncountable. If $x, y \in A$ and $x \ne y$, then

$$d_\infty(x, y) = 1.$$

For each $a \in A$, choose a point $m_a \in B(a, \frac{1}{2}) \cap M$. Since

$$B(a, \tfrac{1}{2}) \cap B(b, \tfrac{1}{2}) = \varnothing \quad \text{for } a \ne b,$$

the map $a \mapsto m_a$ is injective. But A is uncountable while M is countable, a contradiction.

Proposition 1.9. *If x_n converges, then it is bounded.*

Definition 1.16. We denote by $c(\mathbb{R})$ the space of convergent sequences. From the above proposition, $c(\mathbb{R}) \subseteq \ell^\infty(\mathbb{R})$.

1.2 Complete Metric Spaces

1.2.1 Cauchy Sequences

Definition 1.17. Given a sequence x_n in X, we say that x_n is Cauchy if and only if $\forall \varepsilon > 0\, \exists N \in \mathbb{N}$ such that $d(x_m, x_n) < \varepsilon\ \forall m, n \geq N$.

Proposition 1.10. *In a metric space (X, d), every convergent sequence is Cauchy, and every Cauchy sequence is bounded.*

Proposition 1.11. *Assume x_n is a Cauchy sequence in a metric space (X, d) with a subsequence converging to a point x_0. Then the sequence x_n converges to x_0.*

Proof. Let $\varepsilon > 0$. Then there exists N_1 such that for all $m, n \geq N_1$ we have $d(x_n, x_m) < \frac{\varepsilon}{2}$. Since $x_{n_k} \to x_0$, there exists N_2 such that for all $k \geq N_2$ we have $d(x_{n_k}, x_0) < \frac{\varepsilon}{2}$. Let $K > N_2$ be such that $n_K \geq N_1$, then for $n \geq N_1$

$$d(x_n, x_0) \leq d(x_n, x_{n_K}) + d(x_{n_K}, x_0) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Hence, $x_n \to x_0$. □

Definition 1.18. We say that (X, d) is complete if and only if every Cauchy sequence converges in X.

Example 1.4.

- $(\mathbb{R}, |\cdot|)$ is complete; this follows from Bolzano Weierstrass.
- $(\mathbb{Q}, |\cdot|)$ is not complete. In fact, by the density of the rationals, there exists a sequence $q_n \in \mathbb{Q}$ that converges to $\sqrt{2}$ in $\mathbb{R}$. Then q_n is Cauchy in $\mathbb{R}$ and so it is Cauchy in $\mathbb{Q}$. But if x_n converges to some $q \in \mathbb{Q}$, then it converges to $q \in \mathbb{R}$ and by uniqueness of limits $q = \sqrt{2} \in \mathbb{Q}$, a contradiction.

Example 1.5. $c(\mathbb{R})$ is closed in $\ell^\infty(\mathbb{R})$.

Solution. Let $(x^n)_{n\geq 1}$ be a sequence in $c(\mathbb{R})$ converging to a sequence $x = (x_1, x_2, \dots)$ in $\ell^\infty(\mathbb{R})$. We write $x^n = (x_{n1}, x_{n2}, \cdots)$. Fix $\varepsilon > 0$. Since $x^n \to x$ in ℓ^∞, there exists $N \in \mathbb{N}$ such that

$$d_\infty(x^N, x) = \sup_{i\in\mathbb{N}} |x_i^N - x_i| < \frac{\varepsilon}{3}.$$

Since $x^N \in c$, it is Cauchy; hence there exists $i_0 \in \mathbb{N}$ such that

$$|x_{Ni} - x_{Nj}| < \frac{\varepsilon}{3} \qquad \forall i, j \geq i_0.$$

Then for all $i, j \geq i_0$,

$$|x_i - x_j| \leq |x_i - x_{Ni}| + |x_{Ni} - x_{Nj}| + |x_{Nj} - x_j| < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon.$$

Thus $x = (x_i)_{i\geq 1}$ is Cauchy in $\mathbb{R}$, and since $\mathbb{R}$ is complete, (x_i) converges. Therefore $x \in c$. We conclude that c is closed in $\ell^\infty(\mathbb{R})$. □

Example 1.6. Some homework problems.

1. Show that $\ell^\infty(\mathbb{R})$ is complete.
2. Show that $C([a,b])$ is complete with respect to the sup norm.
3. Is $P([a,b])$ closed in $C([a,b])$ with respect to the sup norm?

Proposition 1.12. *Let (X,d) be a metric space and let $E \subseteq X$ be complete. Then E is closed in X.*

Proof. Let (x_n) be a sequence in E that converges to $x \in X$. Being convergent, it is Cauchy in E, and so it converges in E by completeness. Hence, by uniqueness of the limit, we have $x \in E$, so E is closed in X. □

Remark 1.3. The converse is not true in general. For example, consider $(\mathbb{Q}, |\cdot|)$. It is closed in itself but is not complete.

Proposition 1.13. *Every closed subset of a complete metric space is complete, i.e., the converse is true if X is complete.*

Proof. Let (X,d) be a complete metric space and let $E \subseteq X$ be closed. Consider a Cauchy sequence (x_n) in E. Then (x_n) is Cauchy in X, so it converges to some $x \in X$ since X is complete. Since E is closed, $x \in E$, so (x_n) converges in E. □

Example 1.7. $c(\mathbb{R})$ is complete since it is closed in $\ell^\infty(\mathbb{R})$ which is complete, by Examples 1.5 and 1.6.

1.2.2 Completion of Metric Spaces

Lemma 1.1. *If (X,d) is a metric space, then*

$$|d(x,y) - d(x',y')| \leq d(x,x') + d(y,y')$$

Proof. By the triangular inequality $d(x,y) \leq d(x,x') + d(x',y') + d(y',y)$, then

$$d(x,y) - d(x',y') \leq d(x,x') + d(y,y').$$

Reversing the notation and using the symmetric property of d

$$d(x',y') - d(x,y) \leq d(x,x') + d(y,y').$$

This concludes the proof of the lemma. □

Corollary 1.2. *Let (X,d) be a metric space. If x_n, y_n are Cauchy sequences, then $d(x_n,y_n)$ converges in $\mathbb{R}$.*

Proof. Let $d_n = d(x_n,y_n)$ be a sequence in $\mathbb{R}$. Then

$$|d_n - d_m| = |d(x_n,y_n) - d(x_m,y_m)| \leq d(x_n,x_m) + d(y_n,y_m).$$

Since x_n, y_n are Cauchy in X, d_n is Cauchy in $\mathbb{R}$, which by completeness implies that d_n converges. □

Lemma 1.2. *Let (X, d) be a metric space and M a dense subset of X such that every Cauchy sequence in M converges in X. Then X is complete.*

Proof. Let (x_n) be a Cauchy sequence in X. Since M is dense, for every $n \in \mathbb{N}$ there exists $y_n \in M$ such that $d(x_n, y_n) \leq \frac{1}{n}$. Using the triangle inequality,

$$d(y_n, y_m) \leq d(y_n, x_n) + d(x_n, x_m) + d(x_m, y_m) < \frac{1}{n} + d(x_n, x_m) + \frac{1}{m}.$$

Given $\varepsilon > 0$, since x_n is Cauchy, there exists N such that $\frac{1}{N} < \varepsilon/3$ and $d(x_n, x_m) < \varepsilon/3$ for $n, m \geq N$, and so $d(y_n, y_m) < \varepsilon$. We conclude that y_n is Cauchy in M, hence it converges to some $y \in X$.

We show that x_n converges to y. Let $\varepsilon > 0$. There exists N' such that $\frac{1}{N'} < \varepsilon/2$ and $d(y_n, y) < \varepsilon/2$ for $n \geq N'$. Hence for such n,

$$d(x_n, y) \leq d(x_n, y_n) + d(y_n, y) < \frac{1}{n} + d(y_n, y) < \varepsilon.$$

We conclude that x_n converges in X to y, and so X is complete. □

Definition 1.19. A map $\varphi : (X, d_X) \to (Y, d_Y)$ is an isometry if and only if

$$d_Y(\varphi(x), \varphi(y)) = d_X(x, y) \qquad \forall\, x, y \in X.$$

Note that every isometry is injective. If, in addition, ϕ is bijective, we then say that (X, d_X) and (Y, d_Y) are isometric.

Theorem 1.1 (Completion Theorem)**.** *Let (X, d) be a metric space. Then there exists a complete metric space $(\hat{X}, \hat{d})$ and a dense subset M of $\hat{X}$ such that M is isometric to X. $\hat{X}$ is unique up to isometry, and we call it the completion of X.*

Proof. Let C_X be the space of Cauchy sequences in X. We define the relation $\sim$ on C_X as follows

$$x_n \sim y_n \iff \lim_{n\to\infty} d(x_n, y_n) = 0$$

This is indeed an equivalence relation:

- Reflexive: $d(x_n, x_n) = 0$ for all n, so $\lim_{n\to\infty} d(x_n, x_n) = 0$.
- Symmetric: If $x_n \sim y_n$ then since $d(x_n, y_n) = d(y_n, x_n)$ for all n, so
$$0 = \lim_{n\to\infty} d(x_n, y_n) = \lim_{n\to\infty} d(y_n, x_n).$$
- Transitive: If $x_n \sim y_n$ and $y_n \sim z_n$ then $\lim_{n\to\infty} d(x_n, y_n) = 0$ and $\lim_{n\to\infty} d(y_n, z_n) = 0$ and by the triangle inequality we have $d(x_n, z_n) \leq d(x_n, y_n) + d(y_n, z_n) \to 0$, so $x_n \sim z_n$.

Let $\hat{X} = C_X/\sim$ be the set of equivalence classes, so

$$\hat{X} = \{[x_n] : x_n \in C_X\}$$

and define

$$\hat{d}([x_n],[y_n]) = \lim_{n\to\infty} d(x_n, y_n).$$

By Corollary 1.2, $\hat{d}$ exists and is well defined. In fact, if $x_n \sim x'_n$ and $y_n \sim y'_n$ then $|d(x_n, y_n) - d(x'_n, y'_n)| \leq d(x_n, x'_n) + d(y_n, y'_n)$. From Corollary 1.2, $d(x_n, y_n)$ and $d(x'_n, y'_n)$ converge, and since the right-hand side of the inequality goes to zero then $\lim_{n\to\infty} d(x_n, y_n) = \lim_{n\to\infty} d(x'_n, y'_n)$.

Now we show that $\hat{d}$ is a metric on $\hat{X}$:

- $\hat{d}([x_n],[y_n]) \geq 0$ with equality if and only if $\lim_{n\to\infty} d(x_n, y_n) = 0$ if and only if $x_n \sim y_n$ if and only if $[x_n] = [y_n]$.
- $\hat{d}([x_n],[y_n]) = \lim_{n\to\infty} d(x_n, y_n) = \lim_{n\to\infty} d(y_n, x_n) = \hat{d}([y_n],[x_n])$
- Since $d(x_n, z_n) \leq d(x_n, y_n) + d(y_n, z_n)$ for every $x_n, y_n, z_n \in C_X$, letting $n \to \infty$

$$\hat{d}([x_n],[z_n]) \leq \hat{d}([x_n],[y_n]) + \hat{d}([y_n],[z_n]).$$

Now we define $\varphi : X \to \hat{X}$ as follows

$$\varphi(x) = [(x, x, x, x, \ldots)].$$

Notice that

$$\hat{d}(\varphi(x), \varphi(y)) = \lim_{n\to\infty} d(x, y) = d(x, y).$$

Hence φ is an isometry.

We define

$$M = \varphi(X) = \{[(x, x, x, \ldots)] : x \in X\}.$$

We prove that $\overline{M} = \hat{X}$. Take $[x_n] \in \hat{X}$ with (x_n) a Cauchy sequence, we shall show that $B([x_n], \varepsilon) \cap M \neq \emptyset$ for every $\varepsilon > 0$.

Given $\varepsilon > 0$, there exists N such that for $n, m \geq \mathbb{N}$

$$d(x_n, x_m) < \frac{\varepsilon}{2}.$$

Consider the constant sequence (y_n) such that $y_n = x_N$ for every n, so

$$d(x_n, y_n) = d(x_n, x_N) < \frac{\varepsilon}{2} \quad \text{for } n \geq N$$

and

$$\hat{d}([x_n],[y_n]) = \lim_{n\to\infty} d(x_n, y_n) \leq \frac{\varepsilon}{2} < \varepsilon.$$

Hence, $[y_n] = \varphi(x_N) = [(x_N, x_N, \ldots)] \in B([x_n], \varepsilon) \cap M$ and hence $\overline{M} = \hat{X}$.

It remains to prove that $\hat{X}$ is complete. We use Lemma 1.2, take a Cauchy sequence p_{x_n} in M as follows

$$\begin{aligned}
p_{x_1} &= \varphi(x_1) = [(x_1, x_1, x_1, \ldots)] \\
p_{x_2} &= \varphi(x_2) = [(x_2, x_2, x_2, \ldots)] \\
&\vdots \\
p_{x_n} &= \varphi(x_n) = [(x_n, x_n, x_n, \ldots)]
\end{aligned}$$

Consider the sequence $(x_n) = (x_1, x_2, x_3, \ldots)$, we have

$$d(x_n, x_m) = \hat{d}(\varphi(x_n), \varphi(x_m)) = \hat{d}(p_{x_n}, p_{x_m}).$$

Since p_{x_n} is Cauchy in $\hat{X}$ we deduce that (x_n) is Cauchy in X, i.e $(x_n) \in C_X$.

Define then $p = [(x_n)] \in \hat{X}$. Let $\varepsilon > 0$, there exists N such that for $n, k \geq N$ we have $d(x_n, x_k) \leq \frac{\varepsilon}{2}$, and so for $k \geq N$

$$\hat{d}(p, p_{x_k}) = \lim_{n\to\infty} d(x_n, x_k) \leq \frac{\varepsilon}{2} < \varepsilon$$

which shows that p_{x_k} converges to p in $\hat{X}$, so by lemma 1.2 $\hat{X}$ is complete.

Uniqueness is proved in the problem set. □

1.3 Compactness

Definition 1.20. Let (X, τ) be a topological space and $K \subseteq X$. We say that K is compact if and only if every open cover $\mathcal{G} = \{G_\alpha\}_{\alpha\in I}$ of K has a finite subcover.

Example 1.8. $[0, 1)$ is not compact since the open cover $\{(-1, 1 - \frac{1}{n})\}_{n\in\mathbb{N}}$ has no finite subcover.

Example 1.9. Consider a finite subset $A = \{a_1, a_2, a_3, \ldots, a_N\}$ of a toplogical space X and let $\mathcal{G} = \{G_\alpha\}_\alpha$ be an open cover of A, i.e. $A \subseteq \bigcup_{\alpha\in I} G_\alpha$. For each i, there exists $\alpha_i \in I$ such that $a_i \in G_{\alpha_i}$ so $\{G_{\alpha_1}, G_{\alpha_2}, \ldots, G_{\alpha_N}\}$ is a finite subcover of $\mathcal{G}$, deducing that A is compact.

Proposition 1.14. *In a topological space (X, τ), every closed subset of a compact set is compact.*

Proof. Consider a compact set K and a closed subset E of K. Let $\mathcal{G} = \{G_\alpha\}_{\alpha\in I}$ be an open cover of E, so $E \subseteq \bigcup_{\alpha\in I} G_\alpha$ then

$$K \subset E^c \cup \bigcup_{\alpha\in I} G_\alpha.$$

Since E is closed, E^c is open so $\mathcal{G} \cup \{E^c\}$ is an open cover of K. Then there exists a finite subcover $\{G_{\alpha_1}, G_{\alpha_2}, \ldots, G_{\alpha_n}, E^c\}$ of K. Therefore,

$$E = E \cap K \subseteq \bigcup_{i=1}^{n} G_{\alpha_i}$$

deducing that $\{G_{\alpha_1}, G_{\alpha_2}, \ldots, G_{\alpha_n}\}$ is a finite subcover of $\mathcal{G}$ so E is compact. □

Proposition 1.15. *If (X, τ) is a Hausdorff space and $K \subseteq X$ is compact then K is closed.*

Proof. We show that K^c is open. Take $q \in K^c$. Since X is Hausdorff, for every $p \in K$ there exist a neighborhood U_p of p and a neighborhood V_p^q of q such that $U_p \cap V_p^q = \emptyset$.

$\mathcal{G} = \{U_p\}_{p\in K}$ is then an open cover of the compact set K, so it has a finite subcover, i.e.

$$K \subseteq U_{p_1} \cup U_{p_2} \cup \ldots \cup U_{p_n}.$$

Let $V^q = V_{p_1}^q \cap V_{p_2}^q \cap \ldots \cap V_{p_n}^q$. V^q is an open neighborhood of q, and

$$V^q \cap K \subseteq V^q \cap (U_{p_1} \cup U_{p_2} \cup \ldots \cup U_{p_n}) = \emptyset.$$

Then $V^q \subseteq K^c$, which shows that K^c is open so K is closed. □

Proposition 1.16. *In a metric space (X, d), if $K \subseteq X$ is compact then K is closed and bounded.*

Example 1.10. Consider again $\ell^\infty(\mathbb{R})$ with d_∞, and $A = \{0,1\}^{\mathbb{N}}$. Clearly A is bounded. In fact, taking $\mathbf{0} = (0, 0, 0, \dots)$, we have $d(x, \mathbf{0}) \leq 1$ for every $x \in A$. A is closed since, if $y = (y_1, y_2, \cdots)$ in A^c there exists $y_i \notin \{0,1\}$ and there exists $\delta > 0$ such that $(y_i - \delta, y_i + \delta) \cap \{0,1\} = \emptyset$. We get that $B(y, \delta) \subseteq A^c$ so A^c is open, which shows that A is closed.

Although A is closed and bounded, A is not compact. In fact, let $\mathcal{G} = \{B(x, \frac{1}{2}) : x \in A\}$. Notice that $B(x, \frac{1}{2}) \cap A = \{x\}$ but since A is infinite, $\mathcal{G}$ has no finite subcover.

1.3.1 Totally Bounded Sets

Definition 1.21. Let (X, d) be a metric space and $K \subseteq X$. We say that K is totally bounded if and only if $\forall \varepsilon > 0$ there exists $x_1, \dots, x_n \in K$ such that $K \subseteq \bigcup_{i=1}^n B(x_i, \varepsilon)$.

Example 1.11. Following the same reasoning in Example 1.10, $A = \{0,1\}^{\mathbb{N}}$ is not totally bounded in $\ell^\infty(\mathbb{R})$.

Remark 1.4. It is easy to check that every compact set K in a metric space X is totally bounded. In fact, given $\varepsilon > 0$, cover K with balls centered at points of K and radius ε. By compactness, the constructed cover has a finite subcover.

Proposition 1.17. *If K is totally bounded, then K is bounded.*

Proof. Let $\varepsilon = 1$. Then there exists $x_1, \dots, x_n \in K$ such that $K \subseteq \bigcup_{i=1}^n B(x_i, 1)$. For every $x \in K$ there exists i such that $d(x, x_i) < 1$ then $d(x, x_1) \leq d(x, x_i) + d(x_i, x_1) < 1 + d(x_i, x_1)$. Let $M = 1 + \max\{d(x_i, x_1) : i = 1, \dots, n\}$ then $d(x, x_1) < M$ for all $x \in K$ so K is bounded. □

Proposition 1.18. *A subset of a totally bounded set is totally bounded.*

Proof. Let K be totally bounded and $E \subseteq K$. Fix $\varepsilon > 0$. Then we have $x_1, x_2, \cdots, x_n \in K$ s.t.

$$K \subseteq B(x_1, \varepsilon/2) \cup \cdots \cup B(x_n, \varepsilon/2)$$

Pick $y_1, \cdots, y_k$ with $y_i \in B(x_i, \varepsilon/2) \cap E$ (skipping the empty ones). Then we have that

$$E \subseteq B(y_1, \varepsilon) \cup \cdots \cup B(y_k, \varepsilon)$$

□

Proposition 1.19. *In $(\mathbb{R}^n, |\cdot|)$, K is totally bounded if and only if K is bounded.*

Proof. The forward direction is proved for general metric spaces. To prove the converse, assume K is bounded. Then there exists a closed cube in $\mathbb{R}^n$ such that $K \subseteq Q$. Q is compact, so it is totally bounded, and hence $K \subseteq Q$ is totally bounded. □

1.3.2 Classification of compact sets

We show the following theorem that classifies compactness in general metric spaces.

Theorem 1.2. *Let (X, d) be a metric space and $K \subseteq X$. Then the following are equivalent:*

1. *K is compact*
2. *Every infinite subset E of K has a limit point in K (Point compactness).*
3. *Every sequence in K has a convergent subsequence with limit in K (sequential compactness).*
4. *K is complete and totally bounded.*

Proof.

- (1) $\Rightarrow$ (2). If $E \subseteq K$ is infinite, then E has no limit point in K. Then for every $x \in K$ there exists an open neighborhood U_x of x such that $U_x \setminus \{x\} \cap E = \emptyset$. Then $\{U_x\}_{x \in K}$ is an open cover of K. Since K is compact, there exists a finite subcover $\{U_{x_1}, \ldots, U_{x_n}\}$ of K. Then
$$E = E \cap K \subseteq \bigcup_{i=1}^{n} E \cap U_{x_i} \subseteq \{x_1, \cdots, x_n\},$$
a contradiction.

- (2) $\Rightarrow$ (3). Let x_n be a sequence in K. Define the set $E = \{x_n, n \in \mathbb{N}\}$. If E is finite, then there exists $n_1 < n_2 < \ldots$ and $x_0 \in E$ such that $x_{n_k} = x_0$ for all k, so $x_{n_k} \to x_0$. If E is infinite, then by (2) there exists a limit point $x_0 \in K$ of E. Let $\varepsilon = 1$, there exists n_1 such that $x_{n_1} \in B(x_0, 1) \cap E \neq \emptyset$. Having $x_{n_{k-1}}$, $B(x_0, \frac{1}{k}) \cap E \neq \emptyset$; in fact, it is infinite. Then there exists $n_k > n_{k-1}$ such that $x_{n_k} \in B(x_0, \frac{1}{k}) \cap E$. The sequence x_{n_k} then converges to x_0.

- (3) $\Rightarrow$ (4).

  Completeness. Let (x_n) be a Cauchy sequence in K. Since K is sequentially compact, (x_n) has a convergent subsequence $x_{n_k} \to x_0 \in K$. Hence, by Proposition 1.11, $x_n \to x_0 \in K$. Hence K is complete.

  Total boundedness. Suppose, for contradiction, that K is not totally bounded. Then there exists $\varepsilon_0 > 0$ such that K cannot be covered by finitely many balls of radius ε_0 centered at points of K.

  We construct a sequence (x_n) in K as follows. Pick $x_1 \in K$ arbitrarily. Then take $x_2 \in K \setminus B(x_1, \varepsilon_0)$ and continue recursively. Having $x_1, \ldots, x_n \in K$,
  $$B(x_1, \varepsilon_0) \cup \cdots \cup B(x_n, \varepsilon_0)$$
  does not cover K, and there exists
  $$x_{n+1} \in K \setminus \bigcup_{i=1}^{n} B(x_i, \varepsilon_0).$$
  Notice that for $m \neq n$, $d(x_m, x_n) \geq \varepsilon_0$, so no subsequence of x_n would converge, a contradiction. Hence K is totally bounded.

- (4) $\Rightarrow$ (1). Assume K is not compact. Then there exists $\mathcal{G} = \{G_\alpha\}_{\alpha\in I}$ an open cover of K with no finite subcover. Since K is totally bounded, K can be covered by finitely many balls of radius 1 centered at points of K. Then, there exists $x_1 \in K$ and $B(x_1, 1)$ such that $K_1 = B(x_1, 1) \cap K$ cannot be covered by finitely many G_α. Now, since $K_1 \subseteq K$, K_1 is totally bounded and, as above, there exist $x_2 \in K_1$ and $B(x_2, \frac{1}{2})$ such that $K_2 = B(x_2, \frac{1}{2}) \cap K_1$ cannot be covered by finitely many G_α.

  Recursively, we construct a sequence of nested sets

  $$K =: K_0 \supseteq K_1 \supseteq K_2 \supseteq \cdots$$

  with $K_n = B(x_n, \frac{1}{n}) \cap K_{n-1}$, $x_n \in K_{n-1}$, and K_n cannot be covered by finitely many G_α.

  We show that x_n is Cauchy. Let $m, n > N$. Then $x_n, x_m \in K_{n-1}, K_{m-1} \subseteq K_N \subseteq B(x_N, 1/N)$. Thus,

  $$d(x_n, x_m) \leq d(X_n, x_N) + d(x_m, x_N) < \frac{2}{N}.$$

  For $\varepsilon > 0$, choose N such that $\frac{2}{N} < \varepsilon$, we get that $d(x_n, x_m) < \varepsilon$ for $n, m > N$ which shows that x_n is Cauchy.

  Since K is complete, x_n converges to $x_0 \in K$. Since $\mathcal{G}$ is an open cover of K, there exists $\alpha_0 \in I$ such that $x_0 \in G_{\alpha_0}$. Since G_{α_0} is open, there exists $\delta > 0$ such that $B(x_0, \delta) \subseteq G_{\alpha_0}$. Choose N such that $\frac{1}{N} < \delta/2$, and $d(x_N, x_0) < \delta/2$. Notice that

  $$K_N \subseteq B(x_N, 1/N) \subseteq B(x_0, \delta) \subseteq G_{\alpha_0}.$$

  A contradiction. Hence K is compact.

□

1.4 Arzelà-Ascoli

Proposition 1.20. *Let $f : (X, \tau_X) \to (Y, \tau_Y)$ be continuous and let $K \subseteq X$ be compact. Then $f(K)$ is compact in Y.*

Proof. Use definition. □

Corollary 1.3. *Given a metric space (X, d), a compact set $K \subseteq X$, and $f : X \mapsto \mathbb{R}$, a continuous function. Then, there exists $p, q \in K$ such that*

$$f(p) = \sup_K f(x), \quad f(q) = \inf_K f(x).$$

Definition 1.22. Let X be a compact metric space. We denote by $C(X)$ the space of real-valued continuous functions on X. Define the metric

$$d_\infty(f, g) = \sup_{x\in X} |f(x) - g(x)|.$$

The space $(C(X), d_\infty)$ is complete (because uniformly Cauchy in $\mathbb{R}$ implies uniformly convergent) and is separable (follows from the general Stone-Weierstrass theorem).

In all this section, we consider $C(X)$ with the d_∞ metric

Remark 1.5. Recall that, if X is compact and $f : X \to Y$ is continuous, then f is uniformly continuous, i.e.

$$\forall \varepsilon > 0 \ \exists \delta > 0 \text{ such that } d_Y(f(x), f(y)) < \varepsilon \quad \text{for all } x, y \in X \text{ with } d_X(x, y) < \delta.$$

Definition 1.23. $\mathcal{F} \subseteq C(X)$ is equicontinuous if and only if for every $\varepsilon > 0$ there exists $\delta > 0$ such that $d_X(x, y) < \delta$ implies $|f(x) - f(y)| < \varepsilon$ for all $f \in \mathcal{F}$.

Theorem 1.3 (Arzelà-Ascoli)**.** *Let X be a compact metric space. $\mathcal{F} \subseteq C(X)$ is totally bounded if and only if $\mathcal{F}$ is equicontinuous and bounded.*

Proof. ($\Longrightarrow$) $\mathcal{F}$ is bounded since it is totally bounded.

Let $\varepsilon > 0$. Since $\mathcal{F}$ is totally bounded, there exists $f_1, \dots f_n \in \mathcal{F}$ such that $\mathcal{F} \subseteq B(f_1, \frac{\varepsilon}{3}) \cup \dots \cup B(f_n, \frac{\varepsilon}{3})$. Now $f_i \in C(X)$, so it is uniformly continuous. Thus, there exists $\delta_i > 0$ such that $|f_i(x) - f_i(y)| < \frac{\varepsilon}{3}$ for all $x, y \in X$ with $d_X(x, y) < \delta_i$. Let $\delta = \min\{\delta_1, \dots, \delta_n\}$. Take $f \in \mathcal{F}$ and $x, y \in X$ such that $d(x, y) < \delta$. Then there exists $i_0 \in \{1, \dots, n\}$ such that $f \in B(f_{i_0}, \frac{\varepsilon}{3})$ and so

$$|f(x)-f(y)| \leq |f(x)-f_{i_0}(x)|+|f_{i_0}(x)-f_{i_0}(y)|+|f_{i_0}(y)-f(y)| \leq 2d_\infty(f, f_{i_0})+|f_{i_0}(x)-f_{i_0}(y)| < \varepsilon$$

Therefore, $\mathcal{F}$ is equicontinuous.
($\Longleftarrow$) Let $\varepsilon > 0$. Then there exists δ such that $\forall f \in \mathcal{F}$ we have

$$|f(x) - f(y)| < \frac{\varepsilon}{3} \quad \forall x, y \in X \quad \text{such that} \quad d(x, y) < \delta.$$

Since X is compact, we can cover X with finitely many balls of radius δ, i.e.,

$$X = B(x_1, \delta) \cup B(x_2, \delta) \cup \dots \cup B(x_N, \delta).$$

Define the map

$$\varphi : \mathcal{F} \to (\mathbb{R}^N, |\cdot|)$$

as follows:

$$\varphi(f) = (f(x_1), \dots, f(x_N)).$$

We claim that $\varphi(\mathcal{F})$ is bounded in $\mathbb{R}^N$. In fact, since $\mathcal{F}$ is bounded in $C(X)$, there exists $M > 0$ such that $|f(x)| \leq M$ for all $f \in \mathcal{F}$ and $x \in X$. Then

$$|\varphi(f)| = \sqrt{\sum_{i=1}^{N} |f(x_i)|^2} \leq \sqrt{N} M,$$

and the claim follows.

Therefore, $\varphi(\mathcal{F})$ is totally bounded in $\mathbb{R}^n$ (or simply $\overline{\varphi(\mathcal{F})}$ is compact). Then there exists $f_1, \dots, f_K \in \mathcal{F}$ such that

$$\varphi(\mathcal{F}) \subseteq B(\varphi(f_1), \frac{\varepsilon}{3}) \cup \dots \cup B(\varphi(f_K), \frac{\varepsilon}{3}).$$

We prove that $\mathcal{F} \subseteq B(f_1, \varepsilon) \cup \dots \cup B(f_K, \varepsilon)$. Let $f \in \mathcal{F}$. Then there exists $i_0 \in \{1, \dots, K\}$ such that $\varphi(f) \in B(\varphi(f_{i_0}), \frac{\varepsilon}{3})$ and so for every $j = 1, \cdots, N$

$$|f(x_j) - f_{i_0}(x_j)| \leq |\varphi(f) - \varphi(f_{i_0})| < \varepsilon/3.$$

Let $x \in X$. Then $x \in B(x_{j_0}, \delta)$ for some $j_0 \in \{1, \dots, N\}$. Then

$$|f(x) - f_{i_0}(x)| \leq |f(x) - f(x_{j_0})| + |f(x_{j_0}) - f_{i_0}(x_{j_0})| + |f_{i_0}(x_{j_0}) - f_{i_0}(x)| < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon.$$

Hence,

$$d_\infty(f, f_{i_0}) = \sup_X |f(x) - f_{i_0}(x)| = \max_X |f(x) - f_{i_0}(x)| < \varepsilon$$

so $f \in B(f_{i_0}, \varepsilon)$ and therefore $\mathcal{F} \subseteq B(f_1, \varepsilon) \cup \ldots \cup B(f_K, \varepsilon)$ which shows that $\mathcal{F}$ is totally bounded. □

Since closed subsets of complete sets are complete, and compactness is equivalent to completeness and total boundedness, we get the following result.

Corollary 1.4. *$\mathcal{F} \subseteq C(X)$ is compact if and only if $\mathcal{F}$ is equicontinuous, bounded and closed.*

Lemma 1.3. *If $\mathcal{F} \subseteq C(X)$ is equicontinuous, then $\overline{\mathcal{F}}$ is equicontinuous.*

Proof. Let $\varepsilon > 0$. Then there exists $\delta > 0$ such that $d_X(x, y) < \delta$ implies $|f(x) - f(y)| < \varepsilon/2$ for all $f \in \mathcal{F}$.

Let $g \in \overline{\mathcal{F}}$. There exists a sequence $g_n \in \mathcal{F}$ such that $g_n \to g$ uniformly. We have for $x, y \in X$ with $d(x, y) < \delta$

$$|g_n(x) - g_n(y)| < \varepsilon/2.$$

Letting $n \to \infty$, we get

$$|g(x) - g(y)| \leq \varepsilon/2 < \varepsilon,$$

for every $g \in \mathcal{F}$, which shows that $\overline{\mathcal{F}}$ is equicontinuous. □

Theorem 1.4 (Sequential Arzelà-Ascoli)**.** *Let f_n be a sequence in $C(X)$ such that*

i) f_n is equicontinuous, i.e., for every $\varepsilon > 0$ there exists $\delta > 0$ such that for all $n \in \mathbb{N}$, $|f_n(x) - f_n(y)| < \varepsilon$ for $d(x, y) < \delta$.

ii) f_n is uniformly bounded, i.e., there exists $M > 0$ such that $|f_n(x)| \leq M$ for all $n \in \mathbb{N}$ and all $x \in X$.

Then there exists a subsequence f_{n_k} that converges uniformly to some $f \in C(X)$.

Proof. $\mathcal{F} = \{f_n : n \in \mathbb{N}\}$ is equicontinuous and bounded. Then, from Corollary 1.4, $\overline{\mathcal{F}}$ is closed, bounded and equicontinuous and, by Theorem 1.3 $\overline{\mathcal{F}}$ is compact, hence sequentially compact. Therefore, there exist a subsequence $f_{n_k} \to f$ uniformly for some $f \in \overline{\mathcal{F}}$. □

Example 1.12. For $\alpha \in \mathbb{R}$, define the set $\mathcal{F}_\alpha = \{f \in Lip_1([0, 1]) : f(0) = \alpha\}$ where $Lip_1([0, 1]) = \{f : |f(x) - f(y)| \leq |x - y|, x, y \in [0, 1]\}$. Show that $\mathcal{F}_\alpha$ is compact in $C([0, 1])$, then deduce that if $g \in C([0, 1])$ then there exists a function $f \in Lip_1([0, 1])$ such that $f(0) = g(0)$

$$d_\infty(f, g) = \inf\{d_\infty(h, g) : h \in Lip_1([0, 1]), h(0) = g(0)\}.$$

Solution. Notice that $\mathcal{F}_\alpha$ is closed. In fact, if $f_n \in \mathcal{F}_\alpha$ converges to f, then

$$f(0) = \lim_{n\to\infty} f_n(0) = \alpha.$$

Also

$$|f(x) - f(y)| = \lim_{n\to\infty} |f_n(x) - f_n(y)| \leq |x - y|.$$

Thus, $f \in \mathcal{F}_\alpha$.

Next we prove that $\mathcal{F}_\alpha$ is bounded. In fact, for $f \in \mathcal{F}_\alpha$

$$|f(x)| \leq |f(x) - f(0)| + |f(0)| \leq |x| + |\alpha| \leq 1 + |\alpha|.$$

Finally, $\mathcal{F}_\alpha$ is equicontinuous by using the Lipschitz condition and taking $\delta = \varepsilon$. We conclude from Corollary 1.4 that $\mathcal{F}_\alpha$ is compact.

For the second part, apply sequential compactness to $\mathcal{F}_\alpha$ with $\alpha = g(0)$.

1.5 Baire Category

Definition 1.24. Let (X, τ) be a topological space and $A \subseteq X$. We say that A is nowhere dense if and only if $(\overline{A})^\circ = \emptyset$.

Example 1.13. Some examples are:

- The singletons $\{x\}$ in $(\mathbb{R}, |\cdot|)$ are nowhere dense since $\overline{\{x\}} = \{x\}$ and $\{x\}^\circ = \emptyset$.
- Closed sets with empty interiors are nowhere dense; for example, $\mathbb{N}$ and $\mathbb{Z}$ are nowhere dense in $(\mathbb{R}, |\cdot|)$.
- $\mathbb{Q}$ is not nowhere dense in $\mathbb{R}$ since $\overline{\mathbb{Q}} = \mathbb{R}$ and $\mathbb{R}^\circ = \mathbb{R} \neq \emptyset$.

Proposition 1.21. *A is nowhere dense in X if and only if $X \setminus \overline{A}$ is dense in X.*

Proof. ($\Longrightarrow$) Let $x \in X$ and U_x be an open neighborhood of x. Since A is nowhere dense, $U_x \cap X \setminus \overline{A} \neq \emptyset$ which shows that $\overline{X \setminus \overline{A}} = X$.
($\Longleftarrow$) Similar to above. □

Definition 1.25. Let (X, τ) be a topological space and $A \subseteq X$. We say that A is first category or meagre if and only if A can be written as a countable union of nowhere dense subsets of X. Otherwise, we say that A is second category or non-meagre.

Example 1.14. For example, $\mathbb{Q}$ is first category in $\mathbb{R}$ since $\mathbb{Q} = \bigcup_{q\in\mathbb{Q}} \{q\}$ and $\{q\}$ is nowhere dense in $\mathbb{R}$.

Theorem 1.5 (Baire Category Theorem)**.** *Let (X, d) be a complete metric space. If $G_1, G_2, \ldots$ are open dense subsets of X, then $\bigcap_{n=1}^\infty G_n$ is dense in X. Equivalently, every non-empty open subset of X is second category.*

In particular, if $X = \bigcup_{n=1}^\infty F_n$ with F_n closed then there exists n_0 such that F_{n_0} has non-empty interior.

Question 1.1 (Why equivalently?)**.** ($\Longrightarrow$) Let G be open in X and suppose

$$G = \bigcup_{n=1}^{\infty} A_n \subseteq \bigcup_{n=1}^{\infty} \overline{A_n}$$

with A_n nowhere dense. Then $G_n = X \setminus \overline{A_n}$ is open and dense. Thus,

$$\bigcap_{n=1}^{\infty} X \setminus \overline{A_n} = \left(\bigcup_{n=1}^{\infty} \overline{A_n} \right)^c$$

is dense in X. Contradiction, since $G \cap \left(\bigcup_{n=1}^{\infty} \overline{A_n} \right)^c = \emptyset$.
($\Longleftarrow$) Let $G_1, G_2, \ldots$ be open dense subsets of X; each G_i second category. Suppose that $\bigcap_{n=1}^{\infty} G_n$ is not dense. Then there exists non-empty open set H such that

$$H \subseteq \left(\bigcap_{n=1}^{\infty} G_n \right)^c = \bigcup_{n=1}^{\infty} G_n^c$$

Thus,

$$H = \bigcup_{n=1}^{\infty} G_n^c \cap H.$$

Being open, H is second category, there exists n_0 such that $G_{n_0}^c \cap H$ is not nowhere dense, and so

$$\overline{G_{n_0}^c \cap H} \subseteq \overline{G_{n_0}^c} = G_{n_0}^c$$

has non-empty interior, a contradiction, since G_{n_0} is dense in X.

Proof of BCT. Let $x_0 \in X$ and $\delta > 0$, and set $B_0 = B(x_0, \delta)$. We shall show that

$$B(x_0, \delta) \cap (\cap_{n=1}^{\infty} G_n) \neq \emptyset.$$

Since G_1 is dense in X, $B_0 \cap G_1 \neq \emptyset$. Take $x_1 \in B_0 \cap G_1$. Since G_1 is open, choose $\delta_1 < \frac{\delta}{2}$, such that

$$B_1 = B(x_1, \delta_1) \subseteq \overline{B(x_1, \delta_1)} \subseteq B_0 \cap G_1.$$

Since G_2 is dense in X, $B_1 \cap G_2 \neq \emptyset$. Thus, there exists $x_2 \in B_1 \cap G_2$ and $\delta_2 \leq \frac{\delta}{2^2}$ such that

$$B_2 = B(x_2, \delta_2) \subseteq \overline{B(x_2, \delta_2)} \subseteq B_1 \cap G_2 \subseteq B_0 \cap G_1 \cap G_2.$$

Proceeding inductively, suppose we have x_{n-1}, $\delta_{n-1} < \frac{\delta}{2^{n-1}}$, and $B_{n-1} = B(x_{n-1}, \delta_{n-1})$, since G_n is dense in X, $B_{n-1} \cap G_n \neq \emptyset$. Therefore, there exist $x_n \in B_{n-1} \cap G_n$ and $\delta_n < \frac{\delta}{2^n}$ such that,

$$B_n = B(x_n, \delta_n) \subseteq \overline{B(x_n, \delta_n)} \subseteq B_{n-1} \cap G_n.$$

Consequently,

$$B_n \subseteq B_0 \cap G_1 \cap \cdots \cap G_n.$$

We prove that (x_n) is Cauchy. Let $n, m > N$. Since the balls are nested,

$$x_n, x_m \in B_N = B(x_N, \delta_N).$$

Therefore, by the triangle inequality,

$$d(x_n, x_m) \leq d(x_n, x_N) + d(x_m, x_N) < 2\delta_N < \frac{\delta}{2^{N-1}}.$$

Given $\varepsilon > 0$, choose $N \in \mathbb{N}$ sufficiently large such that $\frac{\delta}{2^{N-1}} < \varepsilon$. Then, for all $n, m > N$,

$$d(x_n, x_m) < \varepsilon.$$

Hence, (x_n) is a Cauchy sequence.

Since X is complete, x_n converges to an element $z \in X$. Fix m. For all $n > m$, we have $x_n \in \overline{B_m}$, and since $\overline{B_m}$ is closed

$$z \in \overline{B_m} \subseteq B_0 \cap G_1 \cap \cdots \cap G_m \qquad \forall m.$$

This implies

$$z \in B_0 \cap \left(\bigcap_{n=1}^{\infty} G_n \right).$$

We have shown that for every $x_0 \in X$ and $\delta > 0$ $B(x_0, \delta) \cap \bigcap_{n=1}^{\infty} G_n \neq \emptyset$, so $\bigcap_{n=1}^{\infty} G_n$ is dense in X. □

Chapter 2

Normed Spaces

2.1 Norms

Definition 2.1. Given a vector space V over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, a norm is a map $\|\cdot\| : V \to \mathbb{R}$ satisfying

i) $\|x\| \geq 0$, with equality if and only if $x = 0$.

ii) $\|\alpha x\| = |\alpha| \cdot \|x\|$, $\alpha \in \mathbb{F}$, $x \in V$.

iii) $\|x + y\| \leq \|x\| + \|y\|$, $\forall x, y \in V$.

Example 2.1. $\ell^\infty(\mathbb{R})$ (the set of bounded sequences) under coordinate-wise addition and scalar multiplication and

$$\|x\|_\infty = \sup_{i \in \mathbb{N}} |x_i| = d_\infty(x, 0).$$

$C(X)$ with X compact and

$$\|f\|_\infty = \sup_{x \in X} |f(x)|.$$

Proposition 2.1. *Every norm induces a metric*

$$d(x, y) = \|x - y\|.$$

This allows us to deal with normed spaces as metric spaces with respect to the induced metric.

Remark 2.1. Given a norm $\|\cdot\|$ on V, we will denote by $(V, \|\cdot\|)$ the metric space V with the corresponding metric induced by $\|\cdot\|$. Notice that, by continuity of the metric, the norm is a continuous map from V to $\mathbb{R}$.

Definition 2.2. Given a vector space V and two norms $\|\cdot\|_a, \|\cdot\|_b$ on V, we say that $\|\cdot\|_a$ and $\|\cdot\|_b$ are equivalent if and only if they induce the same topology.

Proposition 2.2. *Given a vector space V and two norms $\|\cdot\|_a, \|\cdot\|_b$ on V, the following are equivalent:*

i) $\|\cdot\|_a$ and $\|\cdot\|_b$ are equivalent.

ii) $\exists C_1, C_2 > 0$ such that

$$C_2 \|x\|_b \leq \|x\|_a \leq C_1 \|x\|_b, \quad \forall x \in V.$$

iii) $x_n \to x_0$ *in* $(V, \|\cdot\|_a) \iff x_n \to x_0$ *in* $(V, \|\cdot\|_b)$.

Proof. [*i)* $\implies$ *ii)*] $B_{\|\cdot\|_a}(0,1) = \{x \in V,\ \|x\|_a < 1\}$ is open in $(V, \|\cdot\|_a)$, and hence $B_{\|\cdot\|_a}(0,1)$ is open in $(V, \|\cdot\|_b)$. Therefore, 0 is an interior point of $B_{\|\cdot\|_a}(0,1)$ with respect to the $\|\cdot\|_b$ norm, and so there exists $\delta > 0$ such that

$$B_{\|\cdot\|_b}(0,\delta) \subseteq B_{\|\cdot\|_a}(0,1).$$

Now take $x \in V$, $x \neq 0$ (the inequality holds trivially at 0). Then

$$\frac{\delta}{2}\frac{x}{\|x\|_b} \in B_{\|\cdot\|_b}(0,\delta) \subseteq B_{\|\cdot\|_a}(0,1).$$

Hence

$$\left\|\frac{\delta}{2}\frac{x}{\|x\|_b}\right\|_a < 1 \implies \|x\|_a < \frac{2}{\delta}\|x\|_b .$$

Switching $\|\cdot\|_a$ and $\|\cdot\|_b$, we similarly obtain the other inequality.

[*ii)* $\implies$ *iii)*] This follows immediately from the squeeze theorem and the following inequality:

$$C_2 \|x_n - x_0\|_b \leq \|x_n - x_0\|_a \leq C_1 \|x_n - x_0\|_b .$$

[*iii)* $\implies$ *i)*] A set is closed in $(V, \|\cdot\|_a)$ if and only if it is closed in $(V, \|\cdot\|_b)$, by the sequential property of closed sets.

□

Definition 2.3. A complete normed space $(V, \|\cdot\|)$ is called a Banach space.

Example 2.2. The space of polynomial $\mathcal{P}([a,b])$ is not Banach in the sup norm, whereas ℓ^1 and ℓ^∞ are Banach.

2.2 Finite Dimensional Vector Spaces

Definition 2.4. Let V be a finite-dimensional vector space and $S = \{v_1, \cdots, v_n\}$ a basis of V. Take $x \in V$. Then there exists unite $r_1, r_2, \cdots, r_n$ such that

$$x = r_1 v_1 + \cdots + r_n v_n.$$

For such a basis S, we define

$$\|x\|_\infty := \max_{1 \leq i \leq n} |r_i|.$$

Proposition 2.3. *Given a finite-dimensional vector space V and a basis $S = \{v_1, \cdots, v_n\}$ of V, the following hold:*

1. $(V, \|\cdot\|_\infty)$ *is Banach.*
2. *The set* $Q_M = \{x \in V :\ \|x\|_\infty \leq M\}$ *is compact.*
3. $K \subseteq (V, \|\cdot\|_\infty)$ *is compact if and only if K is closed and bounded.*

Proof. 1. Let x_k be a Cauchy sequence in V. Then $x_k = r_{k1}v_1 + \cdots + r_{kn}v_n$. For each i, the sequence $(r_{ki})_k$ is Cauchy in $\mathbb{R}$ since

$$|r_{ki} - r_{\ell i}| \leq \|x_k - x_\ell\|_\infty .$$

Hence, it converges to some r_i. Take the vector $x = r_1v_1 + \cdots + r_nv_n$. We get

$$\begin{aligned}\|x_k - x\|_\infty &= \|(r_{k1} - r_1)v_1 + \cdots + (r_{kn} - r_n)v_n\|_\infty \\ &\leq |r_{k1} - r_1| \, \|v_1\|_\infty + \cdots + |r_{kn} - r_n| \, \|v_n\|_\infty \to 0,\end{aligned}$$

concluding that x_k converges to x, and so $(V, \|\cdot\|_\infty)$ is Banach.

2. Take a sequence $x_k = r_{k1}v_1 + \cdots + r_{kn}v_n$ such that $\|x_k\|_\infty \leq M$. Then, for fixed i, $|r_{ki}| \leq M$ for every k; i.e., the sequences $(r_{ki})_k$ are bounded in $\mathbb{R}$. Applying Bolzano–Weierstrass n times, we construct a subsequence $r_{k_\ell i}$ that converges for every i. Let $r_i = \lim_{\ell\to\infty} r_{k_\ell i}$. Then, as in the proof of 1., x_{k_ℓ} converges to x, concluding that $\{x \in V \colon \|x\|_\infty \leq M\}$ is sequentially compact and hence compact.

3. ($\Longrightarrow$) is always true. ($\Longleftarrow$) Since K is bounded, there exists M such that

$$K \subseteq \{x \in V \colon \|x\|_\infty \leq M\} .$$

Since K is a closed subset of a compact set, we obtain that K is compact.

□

Theorem 2.1. *All norms on a finite-dimensional vector space V are equivalent.*

Remark 2.2. Note that the converse is not true in general. Take $P([a, b])$ as a subspace of $C([a, b])$ with the sup norm.

Proof. Take $S = \{v_1, \cdots, v_n\}$, a basis of V, and define

$$\|x\|_\infty = \max_{1\leq i\leq N} |r_i|.$$

Let $\|\cdot\|_a$ be a norm on V. It is enough to show that $\|\cdot\|_a$ is equivalent to $\|\cdot\|_\infty$.

We have

$$\begin{aligned}\|x\|_a &= \|r_1v_1 + \cdots + r_nv_n\|_a \\ &\leq |r_1| \, \|v_1\|_a + \cdots + |r_n| \, \|v_n\|_a \\ &\leq (\|v_1\|_a + \cdots + \|v_n\|_a) \, \|x\|_\infty .\end{aligned}$$

Letting $C_1 = \|v_1\|_a + \cdots + \|v_n\|_a$, we get

$$\|x\|_a \leq C_1\|x\|_\infty.$$

From the above inequality, the map $\varphi \colon (V, \|\cdot\|_\infty) \to \mathbb{R}$ given by $\varphi(x) = \|x\|_a$ is continuous. In fact, if $x_n \to x_0$ in $(V, \|\cdot\|_\infty)$, then

$$|\varphi(x_n) - \varphi(x_0)| = \big| \, \|x_n\| - \|x_0\| \, \big| \leq \|x_n - x_0\|_a \leq C_1 \, \|x_n - x_0\|_\infty \to 0.$$

Now, let $K = \{x \in V \colon \|x\|_\infty = 1\}$. Since K is compact (closed and bounded) in $(V, \|\cdot\|_\infty)$, $\varphi(K)$ is compact in $\mathbb{R}$. Thus, $m = \inf \varphi(K)$ is attained at some $x_0 \in K$. Notice that $\|x_0\|_\infty = 1$, so $x_0 \neq 0$, and hence

$$m = \|x_0\|_a > 0.$$

Let $x \in V$ with $x \neq 0$. Then $\dfrac{x}{\|x\|_\infty} \in K$, and so

$$\varphi\left(\frac{x}{\|x\|_\infty}\right) = \left\|\frac{x}{\|x\|_\infty}\right\|_a \geq m.$$

We hence obtain

$$\|x\|_a \geq m \|x\|_\infty .$$

□

Corollary 2.1. *Since all norms on a finite-dimensional vector space are equivalent, the following properties are independent of the choice of norm:*

- *Every finite-dimensional subspace of a normed space is complete (and hence closed).*
- *Compactness in finite-dimensional spaces is equivalent to being closed and bounded.*
- *Every bounded sequence in a finite-dimensional space has a convergent subsequence.*

We show next that the characterization of compact sets as closed and bounded holds only in finite-dimensional spaces. We first need the following lemma.

Lemma 2.1 (Riesz Lemma)**.** *Given a normed space $(V, \|\cdot\|)$ and a proper closed subspace $\mathcal{M}$ of V, for every $0 < \alpha < 1$ there exists $x \in V$ s.t. $\|x\| = 1$ and*

$$d(x, \mathcal{M}) \geq \alpha.$$

Remark 2.3. In the particular case where $\mathcal{M}$ is finite-dimensional, the distance $d(x, \mathcal{M})$ is attained for every $x \in V$. Therefore, in that case, there exists $x \in V$ such that $\|x\| = 1$ and $d(x, \mathcal{M}) = 1$.

Proof. Take $x_0 \notin \mathcal{M}$. Since $\mathcal{M}$ is closed, $d(x_0, \mathcal{M}) > 0$. Fix $0 < \alpha < 1$. There exists $m_0 \in \mathcal{M}$ s.t.

$$d(x_0, \mathcal{M}) \leq \|x_0 - m_0\| < \frac{d(x_0, \mathcal{M})}{\alpha}.$$

Define $x = \frac{x_0 - m_0}{\|x_0 - m_0\|}$. Note that $\|x\| = 1$, and for every $m \in \mathcal{M}$,

$$\|x - m\| = \left\|\frac{x_0 - m_0}{\|x_0 - m_0\|} - m\right\| = \frac{\|x_0 - (m_0 + m\|x_0 - m_0\|)\|}{\|x_0 - m_0\|}.$$

But $m_0 + m\|x_0 - m_0\| \in \mathcal{M}$, so

$$\|x - m\| \geq \frac{d(x_0, M)}{\|x_0 - m_0\|} > \alpha.$$

Therefore,

$$d(x, \mathcal{M}) = \inf_{m \in \mathcal{M}} \|x - m\| \geq \alpha.$$

□

Theorem 2.2. *Let $(V, \|\cdot\|)$ be a normed space. The unit sphere in V is compact if and only if V is finite-dimensional.*

Proof. ($\Longleftarrow$) Always true.

($\Longrightarrow$) Assume that V is infinite-dimensional. Then we can construct

$$E_1 = \operatorname{Span}\{v_1\}, \quad E_2 = \operatorname{Span}\{v_1, v_2\}, \quad \cdots,$$

where the E_i are finite-dimensional and $\{v_i\}_i$ is linearly independent. Then

$$E_1 \subsetneq E_2 \subsetneq E_3 \subsetneq \cdots,$$

and, being finite-dimensional, the E_i are closed.

Fix some $u_1 \in E_1$ with $\|u_1\| = 1$. Having constructed $u_n \in E_n$ with $\|u_n\| = 1$, by Riesz's Lemma there exists $u_{n+1} \in E_{n+1}$ such that $\|u_{n+1}\| = 1$ and

$$d(u_{n+1}, E_n) \geq 1/2.$$

We obtain as sequence $(u_n)_n$ on the unit sphere such that for $m \neq n$ we have

$$\|u_m - u_n\| \geq 1/2.$$

Therefore, (u_n) has no Cauchy (and hence no convergent) subsequence. Thus, the unit sphere is not sequentially compact and hence not compact. □

2.3 $\ell^p(\mathbb{R})$ Spaces

Lemma 2.2. *Given $1 < p, q < \infty$ such that $\frac{1}{p} + \frac{1}{q} = 1$, assume that $x, y \geq 0$ and $xy = 1$. Then*

$$\frac{x^p}{p} + \frac{y^q}{q} \geq 1,$$

with equality if and only if $x = y = 1$.

Proof. Let $f(x, y) = \frac{x^p}{p} + \frac{y^q}{q}$. We want to minimize $f(x, y)$ subject to the constraint $g(x, y) = xy = 1$. Since, on the closed constraint $xy = 1$, $f(x, 1/x) \to \infty$ as $x \to 0^+$ or $x \to \infty$, f attains a global minimum on $g(x, y) = 1$.

Using Lagrange multipliers, we solve

$$\begin{cases} \nabla f &= \lambda \nabla g, \\ xy &= 1, \end{cases} \implies \begin{cases} x^{p-1} &= \lambda y, \\ y^{q-1} &= \lambda x, \\ xy &= 1. \end{cases}$$

Then we get

$$x^p = \lambda xy = \lambda, \qquad y^q = \lambda xy = \lambda,$$

so $x = \lambda^{1/p}$ and $y = \lambda^{1/q}$. Hence,

$$1 = xy = \lambda^{\frac{1}{p} + \frac{1}{q}} = \lambda.$$

Therefore, $x = y = 1$, and the absolute minimum of f on $g(x, y) = 1$ is $f(1, 1) = 1$. □

Corollary 2.2 (Young inequality)**.** *Given $1 < p, q < \infty$ such that $\frac{1}{p} + \frac{1}{q} = 1$, then*

$$\frac{|x|^p}{p} + \frac{|y|^q}{q} \geq |xy|,$$

with equality if and only if $|x|^p = |y|^q$.

Proof. Without loss of generality, assume $x \neq 0$ and $y \neq 0$. Take

$$X = \frac{|x|}{|xy|^{1/p}} \qquad \text{and} \qquad Y = \frac{|y|}{|xy|^{1/q}}.$$

Notice that

$$XY = \frac{|xy|}{|xy|^{1/p+1/q}} = 1.$$

Applying the lemma, we get

$$1 \leq \frac{X^p}{p} + \frac{Y^q}{q} = \frac{|x|^p}{|xy|p} + \frac{|y|^q}{|xy|q}.$$

Hence,

$$\frac{|x|^p}{p} + \frac{|y|^q}{q} \geq |xy|,$$

with equality if and only if $X = Y = 1$. This is equivalent to

$$|x|^p = |xy| = |y|^q.$$

□

Corollary 2.3 (Hôlder inequality)**.** *Given $1 < p, q < \infty$ such that $\frac{1}{p} + \frac{1}{q} = 1$, and sequences x_n and y_n, then*

$$\sum_{i=1}^{n} |x_i y_i| \leq \left(\sum_{i=1}^{n} |x_i|^p \right)^{1/p} \left(\sum_{i=1}^{n} |y_i|^q \right)^{1/q}.$$

Letting $n \to \infty$ (avoiding $0 \times \infty$), we obtain

$$\sum_{n=1}^{\infty} |x_n y_n| \leq \left(\sum_{n=1}^{\infty} |x_n|^p \right)^{1/p} \left(\sum_{n=1}^{\infty} |y_n|^q \right)^{1/q}.$$

Proof. Without loss of generality, assume that not all x_i and not all y_i are zero. Take

$$X_i = \frac{|x_i|}{\left(\sum_{k=1}^{n} |x_k|^p\right)^{1/p}} \qquad \text{and} \qquad Y_i = \frac{|y_i|}{\left(\sum_{k=1}^{n} |y_k|^q\right)^{1/q}}.$$

From Young's inequality, we have

$$\begin{aligned} X_i Y_i &= \frac{|x_i||y_i|}{\left(\sum_{k=1}^{n} |x_k|^p\right)^{1/p} \left(\sum_{k=1}^{n} |y_k|^q\right)^{1/q}} \\ &\leq \frac{X_i^p}{p} + \frac{Y_i^q}{q} \\ &= \frac{|x_i|^p}{\left(\sum_{k=1}^{n} |x_k|^p\right) p} + \frac{|y_i|^q}{\left(\sum_{k=1}^{n} |y_k|^q\right) q}. \end{aligned}$$

Therefore,

$$\sum_{i=1}^{n} X_i Y_i \leq \frac{1}{p} + \frac{1}{q} = 1.$$

Substituting the definitions of X_i and Y_i gives the desired inequality. □

Corollary 2.4 (Minkowski inequality)**.** *Given sequences x_n and y_n, and $p \geq 1$, then*

$$\left(\sum_{i=1}^{n} |x_i + y_i|^p\right)^{1/p} \leq \left(\sum_{i=1}^{n} |x_i|^p\right)^{1/p} + \left(\sum_{i=1}^{n} |y_i|^p\right)^{1/p}.$$

Letting $n \to \infty$, we obtain

$$\left(\sum_{n=1}^{\infty} |x_n + y_n|^p\right)^{1/p} \leq \left(\sum_{n=1}^{\infty} |x_n|^p\right)^{1/p} + \left(\sum_{n=1}^{\infty} |y_n|^p\right)^{1/p}.$$

Proof. The case $p = 1$ follows from the triangle inequality.

Take $p > 1$ and let q be such that $\frac{1}{p} + \frac{1}{q} = 1$. Then

$$\begin{aligned}
\sum_{i=1}^{n} |x_i + y_i|^p &= \sum_{i=1}^{n} |x_i + y_i| \cdot |x_i + y_i|^{p-1} \\
&\leq \sum_{i=1}^{n} \left(|x_i| + |y_i|\right) |x_i + y_i|^{p-1} \\
&= \sum_{i=1}^{n} |x_i||x_i + y_i|^{p-1} + \sum_{i=1}^{n} |y_i||x_i + y_i|^{p-1} \\
&\leq \left(\sum_{i=1}^{n} |x_i|^p\right)^{1/p} \left(\sum_{i=1}^{n} |x_i + y_i|^{(p-1)q}\right)^{1/q} + \left(\sum_{i=1}^{n} |y_i|^p\right)^{1/p} \left(\sum_{i=1}^{n} |x_i + y_i|^{(p-1)q}\right)^{1/q}.
\end{aligned}$$

We have

$$\frac{1}{p} + \frac{1}{q} = 1 \implies p + q = pq \implies p = q(p-1).$$

Therefore,

$$\sum_{i=1}^{n} |x_i + y_i|^p \leq \left(\sum_{i=1}^{n} |x_i + y_i|^p\right)^{1/q} \left(\left(\sum_{i=1}^{n} |x_i|^p\right)^{1/p} + \left(\sum_{i=1}^{n} |y_i|^p\right)^{1/p}\right).$$

If $x_i + y_i = 0$ for every i, the inequality holds trivially. Otherwise, dividing both sides of the above inequality by $\left(\sum_{i=1}^{n} |x_i + y_i|^p\right)^{1/q}$ gives the desired Minkowski inequality. □

Definition 2.5. Let $1 \leq p < \infty$. We define the space

$$\ell^p(\mathbb{R}) = \left\{ x = (x_1, x_2, \ldots) \colon \sum_{n=1}^{\infty} |x_n|^p < \infty \right\},$$

with the norm

$$\|x\|_p = \left(\sum_{n=1}^{\infty} |x_n|^p\right)^{1/p}.$$

This is indeed a norm by Minkowski's inequality.

As in the case $p = 1$, we can prove the following.

Proposition 2.4. *Given the $\ell^p(\mathbb{R})$ spaces defined above for $1 \leq p < \infty$, we have the following:*

- *$\ell^p(\mathbb{R})$ is Banach and separable.*
- *$K \subseteq \ell^p(\mathbb{R})$ is compact if and only if K is closed and bounded and*

$$\lim_{N\to\infty} \sup_{x\in K} \sum_{n=N+1}^{\infty} |x_n|^p = 0.$$

- *If $1 \leq r < s < \infty$, then*

$$\ell^r(\mathbb{R}) \subsetneq \ell^s(\mathbb{R}).$$

Proof. The first two properties follow very similarly to the case $p = 1$.

Notice that if $r < s$ and $x = (x_1, x_2, \ldots) \in \ell^r(\mathbb{R})$, then $\lim_{n\to\infty} x_n = 0$. Therefore, there exists N such that for $n \geq N$, $|x_n| < 1$, which implies that

$$|x_n|^s \leq |x_n|^r.$$

Hence, by the comparison test, $x \in \ell^s(\mathbb{R})$. Moreover, notice that

$$x_n = \frac{1}{n^{1/r}} \in \ell^s(\mathbb{R}) \setminus \ell^r(\mathbb{R}).$$

□

Remark 2.4. Recall that $\ell^\infty(\mathbb{R})$ is complete but is not separable. Also, if $1 \leq p < \infty$, then $\sum_{n=1}^{\infty} |x_n|^p < \infty$ implies that $x_n \to 0$. Hence, for such p,

$$\ell^p(\mathbb{R}) \subsetneq c_0(\mathbb{R}) \subsetneq \ell^\infty(\mathbb{R}).$$

The first inclusion is strict since

$$x_n = \frac{1}{n^{1/p}} \in c_0(\mathbb{R}) \setminus \ell^p(\mathbb{R}),$$

and the second inclusion is strict since $\mathbf{1} = (1, 1, \cdots) \in \ell^\infty(\mathbb{R}) \setminus c_0(\mathbb{R})$.

2.4 Bounded Linear Transformations

Definition 2.6. Given vector spaces V, W over $\mathbb{F}$, a linear transformation $T : V \mapsto W$ satisfies

$$T(x + \alpha y) = Tx + \alpha Ty \qquad \forall x, y \in V,\ \alpha \in \mathbb{F}.$$

We denote by $\mathcal{L}(V, W)$ the space of linear transformations from V to W.

Remark 2.5. $\mathcal{L}(V,W)$ is a vector space over $\mathbb{F}$. Moreover, when V and W are finite dimensional, $\mathcal{L}(V,W)$ is finite dimensional and

$$\dim \mathcal{L}(V,W) = \dim(V) \cdot \dim(W).$$

Definition 2.7. Take $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$, and let $T \in \mathcal{L}(V,W)$. We say that T is bounded if and only if there exists $M > 0$ such that

$$\|Tx\|_W \leq M \|x\|_V .$$

Equivalently, $\exists M$ such that

$$\|Tx\|_W \leq M, \quad \forall x \text{ such that } \|x\|_V = 1.$$

Proposition 2.5. *Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be normed spaces and $T \in \mathcal{L}(V,W)$. Then*

$$T \text{ bounded} \iff T \text{ is continuous} \iff T \text{ is continuous at } 0.$$

Proof.

- Assume T is bounded. Then there exists $M > 0$ such that $\|Tx\| \leq M \|x\|$. Hence,

  $$\|Tx - Ty\| = \|T(x-y)\| \leq M \|x-y\| .$$

  Thus, T is Lipschitz, so it is continuous.

- Assume T is continuous. Then, trivially, it is continuous at 0.

- If T is continuous at 0, then there exists $\delta > 0$ such that $\|Tx - T0\|_W < 1$ whenever $\|x - 0\|_V < \delta$. This gives

  $$\|Tx\|_W \leq 1$$

  for $\|x\|_V < \delta$. Now take $x \neq 0$. Then

  $$\left\| T\left(\frac{\delta}{2} \frac{x}{\|x\|} \right) \right\| \leq 1.$$

  Thus, we get, for every x,

  $$\|Tx\| \leq \frac{2}{\delta} \|x\| .$$

□

Remark 2.6. For a finite dimensional space $(V, \|\cdot\|_V)$, every linear transformation T is bounded (and hence continuous). In fact, let $S = \{v_1, \cdots, v_n\}$ be a basis of V, and take $x = r_1 v_1 + \cdots + r_n v_n \in V$. Then

$$\|Tx\|_W \leq |r_1| \|Tv_1\|_W + \cdots + |r_n| \|Tv_n\|_W \leq (\|Tv_1\|_W + \cdots + \|Tv_n\|_W) \|x\|_\infty ,$$

where $\|x\|_\infty = \max |r_i|$. Since all norms on V are equivalent, we then have

$$\|Tx\|_W \leq M \|x\|_V .$$

2.4.1 The Operator Norm

Definition 2.8. Given normed spaces $(V, \|\cdot\|)$ and $(W, \|\cdot\|)$, we define $\mathcal{B}(V, W)$ to be the space of bounded linear transformations. This is, in fact, a subspace of $\mathcal{L}(V, W)$. We define the corresponding operator norm by

$$\|T\|_{\mathcal{B}(V,W)} = \sup_{\substack{x\in V \\ \|x\|_V=1}} \|Tx\|_W = \sup_{\substack{x\in V \\ x\neq 0}} \frac{\|Tx\|_W}{\|x\|_V}.$$

Remark 2.7. Notice that for $x \in V$, $x \neq 0$, we have

$$\left\| T\frac{x}{\|x\|_V} \right\|_W \leq \|T\|_{\mathcal{B}(V,W)} .$$

Hence, for every $x \in V$,

$$\|Tx\|_W \leq \|T\|_{\mathcal{B}(V,W)} \cdot \|x\|_V .$$

Also, for every $x, y \in V$,

$$\|Tx - Ty\|_W \leq \|T\|_{\mathcal{B}(V,W)} \cdot \|x - y\|_V.$$

Hence, $\|T\|_{\mathcal{B}(V,W)}$ represents the optimal Lipschitz constant corresponding to T.

Proposition 2.6. *If W is Banach, then $\mathcal{B}(V, W)$ is Banach.*

Proof. Let T_n be a Cauchy sequence in $\mathcal{B}(V, W)$. Notice that

$$\|T_n x - T_m x\| \leq \|T_n - T_m\| \cdot \|x\| .$$

Since T_n is Cauchy in $\mathcal{B}(V, W)$, $T_n x$ is Cauchy in W, which is Banach, and hence it converges. We then define

$$T\colon V \to W, \quad T(x) = \lim_{n\to\infty} T_n(x).$$

Notice that T is linear. In fact,

$$T(x + \alpha y) = \lim_{n\to\infty} T_n(x + \alpha y) = \lim_{n\to\infty} (T_n x + \alpha T_n y) = Tx + \alpha Ty.$$

We next show that T is bounded. In fact, since $\{T_n\}$ is Cauchy in $\mathcal{B}(V, W)$, it is bounded, and so there exists $M > 0$ such that $\|T_n\| \leq M$ for all n. Now,

$$\|T_n x\| \leq \|T_n\| \cdot \|x\| \leq M \|x\| .$$

Letting $n \to \infty$, we get $\|Tx\| \leq M \|x\|$, so $T \in \mathcal{B}(V, W)$.

It remains to prove that $T_n \to T$ in $\mathcal{B}(V, W)$. Given $\varepsilon > 0$, since T_n is Cauchy, there exists N_1 such that

$$\|T_n - T_m\| < \varepsilon/2, \quad \forall n, m \geq N_1.$$

Now fix $x \in V$ such that $\|x\| = 1$. Since $T_n x \to Tx$ in W, there exists $N_2 > N_1$ such that

$$\|T_{N_2} x - Tx\| < \varepsilon/2.$$

Hence, for $n \geq N_1$,

$$\begin{aligned} \|T_n x - Tx\| &\leq \|T_n x - T_{N_2} x\| + \|T_{N_2} x - Tx\| \\ &\leq \|T_n - T_{N_2}\| \cdot \|x\| + \frac{\varepsilon}{2} < \varepsilon. \end{aligned}$$

We conclude that for all $n \geq N_1$ and all $x \in V$ with $\|x\| = 1$, we have

$$\|T_n x - Tx\| < \varepsilon.$$

Therefore,

$$\sup_{\substack{x \in V \\ \|x\|=1}} \|T_n x - Tx\| \leq \varepsilon, \quad \forall n \geq N_1.$$

Hence, $T_n \to T$ in $\mathcal{B}(V, W)$. □

Example 2.3. Define

$$T\colon\ (C([0,1]), \|\cdot\|_\infty) \mapsto \mathbb{R}, \quad T(f) = \int_0^1 f(x)\,dx.$$

Show that $T \in B(C([0,1]), \mathbb{R})$ and find $\|T\|$.

Solution. Clearly, T is linear because integration is linear. Now,

$$|Tf| = \left| \int_0^1 f(x)\,dx \right| \leq \int_0^1 |f(x)|\,dx \leq \|f\|_\infty .$$

Thus, T is bounded and

$$\|T\| = \sup_{\|f\|_\infty = 1} |Tf| \leq 1.$$

On the other hand, $f_0 = 1 \in C[0,1]$, $\|f_0\|_\infty = 1$, and $|Tf_0| = 1$. Therefore, $\|T\| = 1$.

2.4.2 Dual Space

Definition 2.9. Let V be a normed space over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. We define the dual space $V^* = B(V, \mathbb{F})$, which is the space of all bounded linear maps from V to $\mathbb{F}$. We note that such maps are sometimes called bounded linear functionals. The corresponding operator norm is called the dual norm:

$$\|f\|_* = \sup_{\substack{x \in V \\ \|x\|=1}} |f(x)| = \sup_{\substack{x \in V \\ x \neq 0}} \frac{|f(x)|}{\|x\|}.$$

Notice that since $\mathbb{F}$ is Banach, the dual space V^* is always a Banach space.

Remark 2.8. If V is finite dimensional, then $V^* = B(V, \mathbb{F}) = \mathcal{L}(V, \mathbb{F})$, and

$$\dim V^* = \dim V < \infty.$$

Therefore, V^* and V are isomorphic.

Notation. Given $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$, we write $V \cong W$ if and only if V and W are isometrically isomorphic, i.e., there exists an isomorphism $\varphi\colon V \to W$ that is also an isometry.

Theorem 2.3. *We have*

$$(\ell^1)^* \cong \ell^\infty.$$

Moreover, for $1 < p < \infty$,

$$(\ell^p)^* \cong \ell^q,$$

where q *is such that* $\dfrac{1}{p} + \dfrac{1}{q} = 1$.

Remark 2.9. We make a few observations.

- For $p = 2$, we get
$$(\ell^2)^* \cong \ell^2.$$
- One can show that if $V \cong W$, then $V^* \cong W^*$. This implies that, for $1 < p < \infty$,
$$(\ell^p)^{**} \cong (\ell^q)^* \cong \ell^p.$$
- The dual of ℓ^∞ is not ℓ^1, as we will show in the coming chapter, so the above property fails when $p = 1$.

To prove the theorem, we will use the following lemma.

Lemma 2.3. *Let e_i be the sequence whose i-th element is 1 and whose remaining elements are 0. If $1 \le p < \infty$, then for every $x = (x_1, x_2, \cdots) \in \ell^p$, we have*
$$x = \sum_{n=1}^{\infty} x_n e_n = \lim_{N\to\infty} \sum_{n=1}^{N} x_n e_n.$$

Proof. Take $x = (x_1, x_2, \cdots) \in \ell^p$. Notice that
$$\sum_{n=1}^{N} x_n e_n = (x_1, \cdots, x_N, 0, 0, \cdots).$$
Then
$$\left\| x - \sum_{n=1}^{N} x_n e_n \right\|_p = \|(0, 0, \cdots, 0, x_{N+1}, x_{N+2}, \cdots)\|_p = \left(\sum_{n=N+1}^{\infty} |x_n|^p \right)^{1/p} \to 0,$$
since it is the tail of a convergent series. □

Remark 2.10. Above statements is not true for $p = \infty$. For example, taking $x = \mathbf{1} = (1, 1, 1, \cdots)$. Then $\|x - \sum_{n=1}^{N} x_n e_n\|_\infty = 1$, which does not go to zero.

Proof of the duality of ℓ^1. Given $T \in (\ell^1)^*$, by the continuity of T and the above lemma, for every $x \in \ell^1$,
$$Tx = T\left(\sum_{n=1}^{\infty} x_n e_n \right) = \sum_{n=1}^{\infty} x_n T e_n.$$
Define the sequence $b = (Te_1, Te_2, \cdots)$. Notice that
$$|Te_i| \le \|T\|_* \|e_i\|_1 = \|T\|_*.$$
Thus, $b \in \ell^\infty$ and $\|b\|_\infty \le \|T\|_*$. On the other hand,
$$|Tx| = \left| \sum_{n=1}^{\infty} x_n T e_n \right| \le \|b\|_\infty \|x\|_1.$$
Therefore, $\|T\|_* \le \|b\|_\infty$, and hence
$$\|T\|_* = \|b\|_\infty.$$

Using all of the above, we deduce that the map $\psi : (\ell^1)^* \mapsto \ell^\infty$, defined by

$$\psi(T) = (Te_1, Te_2, \cdots),$$

is an isometry. It is also clearly linear. It remains to show that it is surjective.

In fact, for every $b \in \ell^\infty$, consider the map $T_b : \ell^1 \mapsto \mathbb{R}$ defined by

$$T_b(x) = \sum_{n=1}^{\infty} x_n b_n.$$

Clearly, the right-hand side converges, T_b is linear, and

$$|T_b(x)| \leq \|b\|_\infty \|x\|_1,$$

implying that $T_b \in (\ell^1)^*$. Notice that

$$\psi(T_b) = (T_b e_1, T_b e_2, \cdots) = b,$$

showing that ψ is surjective. □

Chapter 3

Hahn-Banach Theorem

3.1 Motivation.

Let $(V, \|\cdot\|)$ be a normed space over $\mathbb{F}$, let $\mathcal{M}$ be a subspace of V, and let $T\colon \mathcal{M} \to \mathbb{F}$ be a bounded linear map. Notice that

$$\|Tx - Ty\| \leq \|T\| \cdot \|x - y\| .$$

Therefore, T is uniformly continuous, and since $\mathbb{F}$ is complete, by the homework, the map

$$\tilde{T}\colon \overline{\mathcal{M}} \to \mathbb{F}, \quad \tilde{T}x = \lim_{n\to\infty} Tx_n,\ x_n \text{ a sequence in } \mathcal{M} \text{ converging to } x$$

is the unique continuous extension of T to $\overline{\mathcal{M}}$. Moreover, it is easy to check that $\overline{\mathcal{M}}$ is a subspace and that $\tilde{T}$ is also linear. Then $\tilde{T} \in (\overline{\mathcal{M}})^*$. Now, by definition,

$$\left\|\tilde{T}\right\|_* = \sup_{\substack{x\in\overline{\mathcal{M}}\\ x\neq 0}} \frac{|\tilde{T}x|}{\|x\|} \geq \sup_{\substack{x\in\mathcal{M}\\ x\neq 0}} \frac{|\tilde{T}x|}{\|x\|} = \sup_{\substack{x\in\mathcal{M}\\ x\neq 0}} \frac{|Tx|}{\|x\|} = \|T\|_* .$$

On the other hand, for $x \in \overline{\mathcal{M}}$, let x_n be a sequence in $\mathcal{M}$ converging to x. Then

$$|Tx_n| \leq \|T\|_* \cdot \|x_n\| .$$

Letting $n \to \infty$, we get

$$|\tilde{T}x| \leq \|T\|_* \cdot \|x\| \implies \left\|\tilde{T}\right\|_* \leq \|T\|_* .$$

We conclude that the defined extension $\tilde{T}$ of T to the closure of $\mathcal{M}$ is unique and satisfies

$$\|\tilde{T}\|_* = \|T\|_*.$$

So far, we have extended T to $\overline{\mathcal{M}}$ using continuity. The question now is whether we can extend T beyond $\overline{\mathcal{M}}$ while preserving its norm. We start by showing that this is always possible when adding one dimension.

Lemma 3.1. *Given a normed space V over $\mathbb{R}$ and a proper subspace $\mathcal{M}$ of V, let*

$$f\colon \mathcal{M} \to \mathbb{R}$$

be a bounded linear map. Then, for $x_0 \notin \mathcal{M}$, there exists an extension

$$F\colon \mathcal{M} \oplus \operatorname{Span}\{x_0\} \to \mathbb{R}$$

such that F is a bounded linear map, $F_{|\mathcal{M}} = f$, and

$$\|F\|_* = \|f\|_*.$$

Proof. For $x \in \mathcal{M} \oplus \operatorname{Span}\{x_0\}$, we can write $x = m + tx_0$, with $m \in \mathcal{M}$ and $t \in \mathbb{R}$. The extension F should be such that $F(m) = f(m)$ for every $m \in \mathcal{M}$. So, by linearity, defining $F(x_0) = c$ to obtain

$$F(x) = F(m + tx_0) = f(m) + tc$$

gives a linear extension of f to $\mathcal{M} \oplus \operatorname{Span} x_0$.

The goal is to define c so that, for every $t \in \mathbb{R}$,

$$|F(m + tx_0)| \leq \|f\|_* \|m + tx_0\| \quad (\star).$$

In that case, F is bounded, and we would obtain $\|F\|_* = \|f\|_*$ as desired. Since this would give us

$$\|F\|_* = \|f\|_*.$$

Clearly, $(*)$ holds for $t = 0$. Note that it is enough to find c such that $(*)$ holds for $t = 1$, that is, to find c so that, for every $m \in \mathcal{M}$,

$$|F(m + x_0)| = |f(m) + c| \leq \|f\|_* \|m + x_0\| \qquad (\star\star).$$

In fact, suppose such c exists. Then, for $t \neq 0$,

$$|F(m + tx_0)| = |t| \left|F\left(\frac{m}{t} + x_0\right)\right| \leq |t| \|f\|_* \left\|\frac{m}{t} + x_0\right\| = \|f\|_* \|m + tx_0\|.$$

Notice that $(\star\star)$ is equivalent to

$$-f(m) - \|f\|_* \|m + x_0\| \leq c \leq \|f\|_* \|m + x_0\| - f(m), \quad \forall m \in \mathcal{M}. \ (\star\star\star)$$

Notice that, for $m, m' \in \mathcal{M}$,

$$f(m') - f(m) = f(m' - m) \leq |f(m' - m)| \leq \|f\|_* \|m' - m\| \leq \|f\|_* \left(\|m' + x_0\| + \|m + x_0\|\right),$$

and so

$$-f(m) - \|f\|_* \|m + x_0\| \leq -f(m') + \|f\|_* \|m' + x_0\|, \quad \forall m, m' \in \mathcal{M}.$$

Hence,

$$\sup_{m \in \mathcal{M}} \left(-f(m) - \|f\|_* \|m + x_0\|\right) \leq \inf_{m \in \mathcal{M}} \left(-f(m) + \|f\|_* \|m + x_0\|\right).$$

And so we can find c_0 satisfying $(\star\star\star)$, and so the map $F : \mathcal{M} \oplus \operatorname{Span}\{x_0\} \mapsto \mathbb{R}$, defined as follows:

$$F(m + tx_0) = f(m) + tc_0,$$

is a bounded continuous extension of f to $\mathcal{M} \oplus \operatorname{Span}\{x_0\}$ satisfying $\|F\|_* = \|f\|_*$. □

Remark 3.1. The goal is to prove that there exists an extension to all of V. From the above lemma, we can always extend by one more dimension. The process in finite dimensions should stop when we reach the dimension of V. However, in infinite dimensions, we need to "repeat" this infinitely many times. The fact that this would give the existence of a "maximal" extension requires an axiomatic argument using Zorn's Lemma.

3.2 Partially Ordered Sets

Definition 3.1. Given a set X, a partial order on X is a relation $\preceq$ on X that is

i) Reflexive: $x \preceq x$.

ii) Antisymmetric: $x \preceq y$ and $y \preceq x \implies x = y$.

iii) Transitive: $x \preceq y$ and $y \preceq z \implies x \preceq z$.

We say that $(X, \preceq)$ is a partially ordered set.

Example 3.1. Given a set X, the power set with inclusion $(\mathcal{P}(X), \subseteq)$ is a partially ordered set.

We can also define the divisibility partial order on $\mathbb{N}$ as follows:

$$a \preceq b \iff a \text{ divides } b.$$

Notice that, $2 \preceq 6$, but 2 and 3 are not comparable.

Definition 3.2. Let $(X, \preceq)$ be a partially ordered set, and let $A \subseteq X$.

- An element $m \in A$ is called a maximal element of A if m does not precede any element of A except itself. That is, if $a \in A$ is such that $m \preceq a$, then $a = m$.
- An element $m \in X$ is called an **upper bound** of A if $a \preceq m$ for every $a \in A$.

Example 3.2. For example, consider $\mathbb{N}$ with divisibility and $A = \{2, 3, 5, 6\} \subseteq \mathbb{N}$. Then 30 is an upper bound of A, and the elements $5, 6$ are maximal elements of A.

Definition 3.3. Given a partially ordered set $(X, \preceq)$, we say that $(X, \preceq)$ is totally ordered if and only if all elements in X are comparable. That is, for all $x, y \in X$, we have $x \preceq y$ or $y \preceq x$.

Definition 3.4. Let $(X, \preceq)$ be a partially ordered set. A chain $\mathcal{C}$ is a totally ordered subset of X.

Theorem 3.1 (Zorn's Lemma)**.** *Let $(X, \preceq)$ be a nonempty partially ordered set such that every chain in X has an upper bound. Then X has a maximal element.*

Remark 3.2. When applying Zorn's lemma, do not forget to prove that the defined set X is actually nonempty.

Notation. In a partially ordered set $(X, \preceq)$, we write $x \prec y$ if and only if $x \preceq y$ and $x \neq y$.

Example 3.3. A nice application of Zorn's Lemma is that every vector space V has a (Hamel) basis.

Proof. The empty set is a basis of the zero vector space. Suppose that $V \neq 0$, and let X be the set of linearly independent subsets of V, ordered by inclusion. First, notice that X is nonempty because, for any $v_0 \neq 0$ in V, we have that $\{v_0\}$ is linearly independent, so $\{v_0\} \in X$.

Let $\mathcal{C} = \{C_\alpha\}_{\alpha \in I}$ be a (non-empty) chain in X, and define

$$A = \bigcup_{\alpha \in I} C_\alpha.$$

We want to show first that $A \in X$, that is, A is linearly independent. Take $v_1, v_2, \cdots, v_n \in A$ with

$$r_1 v_1 + r_2 v_2 + \cdots + r_n v_n = 0.$$

We have $v_i \in C_{\alpha_i}$ for some $\alpha_i \in I$. Since the C_{α_i} are comparable, by reindexing, we can say

$$C_{\alpha_1} \preceq C_{\alpha_2} \preceq \cdots \preceq C_{\alpha_n}.$$

Therefore, $v_1, v_2, \cdots, v_n \in C_{\alpha_n} \in X$, hence

$$r_1 = r_2 = \cdots = r_n = 0.$$

Hence, $A \in X$, and we have $C_\alpha \preceq A$ for every $\alpha \in I$. Thus, A is an upper bound of $\mathcal{C}$.

Hence, by Zorn's lemma, X has a maximal element S. Since $S \in X$, S is a linearly independent subset of V. Assume $\operatorname{Span} S \subsetneq V$. Then there exists a nonzero vector $v_0 \in V \setminus \operatorname{Span} S$. Therefore, the set $S \cup \{v_0\}$ is linearly independent, i.e., $S \cup \{v_0\} \in X$, and $S \prec S \cup \{v_0\}$, a contradiction. We conclude that S is linearly independent and that $\operatorname{Span} S = V$. Hence, S is a basis of V. □

3.3 Hahn Banach extension Theorems

3.3.1 The Case of a Vector Space over $\mathbb{R}$

Theorem 3.2 (Real Case)**.** *Given a normed space V over $\mathbb{R}$, a subspace $\mathcal{M}$ of V, and a bounded linear functional $f : \mathcal{M} \mapsto \mathbb{R}$, there exists a bounded linear functional $F \colon V \mapsto \mathbb{R}$ such that*

$$F_{|\mathcal{M}} = f, \quad \text{and} \quad \|F\|_* = \|f\|_*.$$

Proof. Define the following set

$$X = \left\{ (h, D(h)) \;\middle|\; \begin{array}{l} \mathcal{M} \subseteq D(h) \subseteq V, \quad D(h) \text{ is a subspace of } V, \\ h : D(h) \to \mathbb{R} \text{ is a bounded linear map}, \\ h_{|\mathcal{M}} = f, \quad \|h\|_* = \|f\|_* \end{array} \right\},$$

with the following partial order

$$(h_1, D(h_1)) \preceq (h_2, D(h_2)) \iff D(h_1) \subseteq D(h_2) \text{ and } h_2|_{D(h_1)} = h_1.$$

Notice that $X \neq \varnothing$ since $(f, \mathcal{M}) \in X$. Let $\mathcal{C} = \{(h_\alpha, D(h_\alpha))\}_{\alpha \in I}$ be a nonempty chain in X. Define

$$\mathcal{D} = \bigcup_{\alpha \in I} D(h_\alpha).$$

Notice that since $\mathcal{M} \subseteq D(h_\alpha)$, we have $\mathcal{M} \subseteq \mathcal{D}$. We claim that $\mathcal{D}$ is a subspace of V. In fact, given $x, y \in \mathcal{D}$ and $\lambda \in \mathbb{R}$, we have $x \in D(h_{\alpha_x})$ and $y \in D(h_{\alpha_y})$ for some $\alpha_x, \alpha_y \in I$. But $(h_{\alpha_x}, D(h_{\alpha_x}))$ and $(h_{\alpha_y}, D(h_{\alpha_y}))$ are comparable, say

$$(h_{\alpha_x}, D(h_{\alpha_x})) \preceq (h_{\alpha_y}, D(h_{\alpha_y})).$$

Then $x, y \in D(h_{\alpha_y})$, which is a subspace of V, and hence $x + \lambda y \in D(h_{\alpha_y}) \subseteq \mathcal{D}$.

Next, define $h \colon \mathcal{D} \mapsto \mathbb{R}$ as follows:

$$h|_{D(h_\alpha)} = h_\alpha.$$

Notice the following:

- h is well-defined. In fact, if $x \in D(h_{\alpha_1}) \cap D(h_{\alpha_2})$, without loss of generality, $(h_{\alpha_1}, D(h_{\alpha_1})) \preceq (h_{\alpha_2}, D(h_{\alpha_2}))$, and so $h_{\alpha_2}(x) = h_{\alpha_1}(x)$.
- $h_{|\mathcal{M}} = f$ since $h_{\alpha|\mathcal{M}}$.
- h is linear since h_α is linear.
- h is bounded and $\|h\|_* = \|f\|_*$. Take $x \in \mathcal{D}$. Then $x \in D(h_{\alpha_x})$ for some $\alpha_x \in I$, so

$$|h(x)| = |h_{\alpha_x}(x)| \leq \|h_{\alpha_x}\|_* \|x\| = \|f\|_* \|x\|.$$

  Thus, h is bounded and $\|h\|_* \leq \|f\|_*$. However, since $\mathcal{M} \subseteq \mathcal{D}$ and $h = f$ on $\mathcal{M}$, we have

$$\|h\|_* = \|f\|_*.$$

We conclude that $(h, \mathcal{D}) \in X$ and is an upper bound for $\mathcal{C}$.

We then apply Zorn's Lemma and deduce the existence of a maximal element $(F, D(F)) \in X$. We have $F_{|\mathcal{M}} = f$ and $\|F\|_* = \|f\|_*$. It remains to show that $D(F) = V$. Assume $D(F) \subsetneq V$. Then there exists a nonzero vector $x_0 \in V \setminus D(F)$. From Lemma 3.1, there exists a bounded linear functional $G : D(F) \oplus \operatorname{Span}\{x_0\} \mapsto \mathbb{R}$ such that $G_{|D(F)} = F$ and $\|G\|_* = \|F\|_* = \|f\|_*$. Letting $D(G) = D(F) \oplus \operatorname{Span}\{x_0\}$, we get that $(G, D(G)) \in X$ and $(F, D(F)) \prec (G, D(G))$, a contradiction. □

Lemma 3.2. *Let V be a vector space over $\mathbb{C}$, and let $f : V \to \mathbb{C}$ be a map. Then f is $\mathbb{C}$-linear if and only if there exists an $\mathbb{R}$-linear map $f_1 : V \to \mathbb{R}$ such that*

$$f(x) = f_1(x) - if_1(ix), \qquad \forall x \in V.$$

In this case, $f_1 = \operatorname{Re} f$.

Moreover, if V is a normed space, then f is bounded if and only if f_1 is bounded, and we obtain

$$\|f\|_* = \|f_1\|_*.$$

Proof of ($\Longrightarrow$). Given $f\colon V \mapsto \mathbb{C}$, we can write $f(x) = f_1(x) + if_2(x)$, where $f_1, f_2\colon V \mapsto \mathbb{R}$. Therefore,

$$f_1(x) = \operatorname{Re} f(x) = \frac{f(x) + \overline{f(x)}}{2}.$$

We show that f_1 is $\mathbb{R}$-linear. For $x, y \in V$ and $\alpha \in \mathbb{R}$,

$$\begin{aligned} f_1(x + \alpha y) &= \frac{f(x + \alpha y) + \overline{f(x + \alpha y)}}{2} \\ &= \frac{f(x) + \alpha f(y) + \overline{f(x)} + \overline{\alpha}\,\overline{f(y)}}{2} \\ &= \frac{f(x) + \overline{f(x)}}{2} + \alpha \frac{f(y) + \overline{f(y)}}{2} \\ &= f_1(x) + \alpha f_1(y). \end{aligned}$$

Moreover, since f is linear, we have

$$\begin{aligned} f(ix) &= if(x), \\ f_1(ix) + if_2(ix) &= if_1(x) - f_2(x). \end{aligned}$$

Therefore, $f_2(x) = -f_1(ix)$, so

$$f(x) = f_1(x) - if_1(ix).$$

Assume f is bounded. Then, for $x \in V$,

$$|f_1(x)| \leq |f(x)| \leq \|f\|_* \|x\|.$$

Therefore, f_1 is bounded with $\|f_1\|_* \leq \|f\|_*$.

To prove the other inequality, notice that for $x \in V$, there exists $\theta_x \in \mathbb{R}$ such that $f(x) = |f(x)|e^{i\theta_x}$. Hence,

$$|f(x)| = f(x)e^{-i\theta_x} = f(e^{-i\theta_x}x) = f_1(e^{-i\theta_x}x) + if_2(e^{-i\theta_x}x).$$

Comparing real and imaginary parts, we get

$$|f(x)| = f_1(e^{-i\theta_x}x) \leq \|f_1\|_* \|x\|.$$

So $\|f\|_* \leq \|f_1\|_*$.

We conclude that $\|f\|_* = \|f_1\|_*$.

The proof of the converse is left as an exercise. □

Theorem 3.3 (General Case)**.** *Given a normed space V over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, a subspace $\mathcal{M}$ of V, and a bounded linear functional $f : \mathcal{M} \mapsto \mathbb{F}$, there exists a bounded linear functional $F\colon V \mapsto \mathbb{F}$ such that*

$$F_{|\mathcal{M}} = f, \quad \text{and} \quad \|F\|_* = \|f\|_*.$$

Proof. The case when $\mathbb{F} = \mathbb{R}$ has already been proved.

If $\mathbb{F} = \mathbb{C}$, use the above lemma to write $f(x) = f_1(x) - if_1(ix)$, with $f_1\colon \mathcal{M} \mapsto \mathbb{R}$ bounded. Regarding V and $\mathcal{M}$ as vector spaces over $\mathbb{R}$, we then extend f_1 to $F_1\colon V \mapsto \mathbb{R}$ with

$$\|F_1\|_* = \|f_1\|_* = \|f\|_*.$$

Hence,

$$F(x) = F_1(x) - iF_1(ix)$$

is the desired extension. □

3.3.2 Consequences of Hahn-Banach

Corollary 3.1. *Given a normed space V and $x_0 \in V$ with $x_0 \neq 0$, there exists $f \in V^*$ such that $\|f\|_* = 1$ and $f(x_0) = \|x_0\|$.*

Proof. Take $\mathcal{M} = \operatorname{Span}\{x_0\}$ and the linear map $g\colon \mathcal{M} \mapsto \mathbb{F}$ defined by

$$g(tx_0) = t\|x_0\|.$$

For $x = tx_0 \in \mathcal{M}$,

$$|g(x)| = |t\|x_0\|| = \|tx_0\| = \|x\|.$$

Therefore,

$$\|g\|_* = 1.$$

Applying Hahn-Banach, we construct an extension f of g to V with $\|f\|_* = \|g\|_* = 1$. We have $f_{|\mathcal{M}} = g$, so $f(x_0) = g(x_0) = \|x_0\|$. □

Corollary 3.2. *If $V \neq \{0\}$, then $V^* \neq \{0\}$.*

Proof. This is an immediate consequence of Corollary 3.1. □

Corollary 3.3. *Given a nonzero normed space V and $x_0 \in V$, we have*

$$\|x_0\| = \sup_{\substack{f \in V^* \\ \|f\|_* = 1}} |f(x_0)| = \max_{\substack{f \in V^* \\ \|f\|_* = 1}} |f(x_0)|.$$

Proof. Let $f \in V^*$ be such that $\|f\|_* = 1$. Then

$$|f(x_0)| \leq \|f\|_* \|x_0\| = \|x_0\|,$$

and so

$$\sup_{\substack{f \in V^* \\ \|f\|_* = 1}} |f(x_0)| \leq \|x_0\|.$$

By Corollary 3.1, the supremum is attained when $x_0 \neq 0$. The result is trivial for $x_0 = 0$. □

Corollary 3.4 (Point-Point Separation)**.** *Given $x \neq y$ in a normed space V, there exists $f \in V^*$ such that $f(x) \neq f(y)$.*

Proof. Since $x \neq y$, we have $x - y \neq 0$ in V. Thus, by Corollary 3.1, there exists $f \in V^*$ such that

$$0 \neq \|x - y\| = f(x - y) = f(x) - f(y),$$

as desired. □

Corollary 3.5 (Closed Subspace-Point Separation)**.** *Given a normed space V over $\mathbb{F}$, a proper closed subspace $\mathcal{M}$ of V, and $x_0 \notin \mathcal{M}$, there exists $f \in V^*$ such that $f_{|\mathcal{M}} = 0$ and $f(x_0) = 1$.*

Proof. Define the linear functional $g \colon \mathcal{M} \oplus \operatorname{Span}\{x_0\} \mapsto \mathbb{F}$ as follows:

$$g(m + tx_0) = t.$$

We show that g is bounded. In fact, take $x = m + tx_0$, with $m \in \mathcal{M}$ and $t \in \mathbb{F}$, such that $\|x\| = 1$. If $t = 0$, then $g(x) = 0$. Otherwise, notice that

$$1 = \|m + tx_0\| = |t| \cdot \left| \frac{m}{t} + x_0 \right| \geq |t| dist(x_0, \mathcal{M}).$$

Hence, $\|m + tx_0\| = 1$ implies that

$$|g(m + tx_0)| = |t| \leq \frac{1}{d(x_0, \mathcal{M})}.$$

Thus, g is bounded and $g_{|\mathcal{M}} = 0$. Applying Hahn-Banach, we obtain the desired functional f. □

Example 3.4. Let $x \in V$ be such that $f(x) = 0$ for every $f \in V^*$. Then $x = 0$; otherwise, we obtain a contradiction to Corollary 3.1.

Example 3.5. We define the map $\varphi\colon (\ell^1) \mapsto (\ell^\infty)^*$ as follows:

$$\varphi(x)\colon b \mapsto \sum_{n=1}^{\infty} b_n x_n,$$

with $b \in \ell^\infty$. Then φ is not surjective.

Consider the closed subspace c_0 of ℓ^∞ and the sequence $\mathbf{1} = (1, 1, \cdots) \in \ell^\infty \setminus c_0$. By the separation corollary, there exists $T \in (\ell^\infty)^*$ such that $T_{|c_0} = 0$ and $T(\mathbf{1}) = 1$. Suppose that there exists $x \in \ell^1$ such that $T = \varphi(x)$. Then

$$Tb = \sum_{n=1}^{\infty} b_n x_n, \quad \forall b \in \ell^\infty.$$

Since $e_i \in c_0$,

$$0 = Te_i = x_i.$$

Thus, $x = 0$, hence $T = 0$, a contradiction.

Example 3.6. Let V and W be normed spaces, with $V \neq \{0\}$. Then $B(V, W)$ is Banach if and only if W is Banach.

We have already shown $\Longleftarrow$. For the forward direction, take $v_0 \in V$ such that $\|v_0\| = 1$. Then there exists $f \in V^*$ with $\|f\|_* = 1$ and $f(v_0) = \|v_0\| = 1$.

For $w \in W$, define $T_w\colon V \mapsto W$ as follows:

$$T_w x = f(x)w.$$

The map T_w is linear since f is linear, and

$$\|T_w x\|_W = |f(x)| \, \|w\| \leq \|x\|_V \, \|w\|_W .$$

Thus, $T_w \in \mathcal{B}(V, W)$ and

$$\|T_w\| \leq \|w\| .$$

However,

$$\|T_w v_0\| = |f(v_0)| \, \|w\| = \|w\| ,$$

and so

$$\|T_w\| = \|w\| .$$

Consider a Cauchy sequence (w_n) in W. Then (T_{w_n}) is Cauchy in $\mathcal{B}(V, W)$, so T_{w_n} converges to some $T \in \mathcal{B}(V, W)$. Take $w_0 = Tv_0$. Notice that $T_w(x) = f(x)Tv_0$ Notice that

$$\|w_n - w_0\| = \|T_{w_n} v_0 - Tv_0\| \leq \|T_{w_n} - T\|$$

Therefore, $w_n \to w$.

Proposition 3.1. *Let V be a normed space. If V^* is separable, then V is separable.*

Remark 3.3. Note that the converse is not true, since ℓ^1 is separable and $(\ell^1)^* \cong \ell^\infty$ is not. Also, this implies that $(\ell^\infty)^*$ is not isomorphic to ℓ^1.

Proof. Without loss of generality, assume $V \neq \{0\}$. Let $\{f_n\}$ be a countable dense subset of V^*. Since

$$\|f_n\|_* = \sup_{\substack{x \in V \\ \|x\|=1}} |f_n(x)|,$$

there exists $x_n \in V$ with $\|x_n\| = 1$ and

$$\frac{\|f_n\|_*}{2} \leq |f_n(x_n)| \leq \|f_n\|_*.$$

Define the set $\mathcal{M} = \operatorname{Span}\{x_n\}$. We claim that $\overline{\mathcal{M}} = V$. Suppose not. Then there exists $x_0 \in V$ such that $x_0 \notin \overline{\mathcal{M}}$. By the separation corollary, there exists $f \in V^*$ with $f(x_0) = 1$ and $f_{|\overline{\mathcal{M}}} = 0$.

We have $f(x_n) = 0$ for every n. We show that this implies $f = 0$, a contradiction since $f(x_0) = 1$. Let $\varepsilon > 0$, and choose N such that

$$\|f_N - f\|_* < \varepsilon/3.$$

Then, for $\|x\| = 1$,

$$\begin{aligned}
|f(x)| &\leq |f(x) - f_N(x)| + |f_N(x)| \\
&\leq \|f - f_N\|_* + \|f_N\|_* \\
&\leq \|f - f_N\|_* + 2|f_N(x_N)| \\
&= \|f - f_N\|_* + 2|f_N(x_N) - f(x_N)| \\
&\leq \|f - f_N\|_* + 2\|f - f_N\|_* \\
&< \varepsilon.
\end{aligned}$$

Therefore, $|f(x)| = 0$, which implies $\|f\|_* = 0$. Hence $f = 0$, a contradiction.

Now,

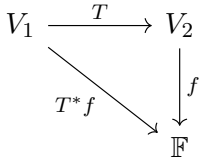
$$\overline{\operatorname{Span}\{x_n\}} = V \implies \overline{\operatorname{Span}_{\mathbb{Q}}\{x_n\}} = V$$

(or, in the complex case, $\overline{\operatorname{Span}_{\mathbb{Q}+i\mathbb{Q}}\{x_n\}} = V$), so V is separable. □

3.4 The Adjoint Linear Transformation

Definition 3.5. Given vector spaces V_1, V_2 and $T \in \mathcal{B}(V_1, V_2)$, we define the adjoint of T to be the map $T^*\colon V_2^* \mapsto V_1^*$ defined by $T^* f = f \circ T$, i.e.,

$$(T^* f, x)_{V_1^*, V_1} = (f, Tx)_{V_2^*, V_2}.$$



This operator corresponds to the transpose of a matrix in finite dimensions, as discussed in Kreyszig, p. 234.

We make the following observations:

- $T^* \in \mathcal{L}(V_2^*, V_1^*)$. In fact,

$$T^*(f + \alpha g) = (f + \alpha g) \circ T = f \circ T + \alpha g \circ T = T^* f + \alpha T^* g.$$

- $T^* \in \mathcal{B}(V_2^*, V_1^*)$ and $\|T^*\| = \|T\|$. In fact, for $x \in V_1$,

$$|(T^* f)(x)| = |f(Tx)| \leq \|f\|_* \|Tx\| \leq \|f\|_* \cdot \|T\| \cdot \|x\|.$$

  Hence, $\|T^* f\|_* \leq \|f\|_* \|T\|$, and so T^* is bounded and

$$\|T^*\| \leq \|T\|.$$

  To prove the reverse inequality, we use Hahn-Banach. In fact, if $Tx \neq 0$, there exists $f \in V_2^*$ with $\|f\|_* = 1$ such that

$$\|Tx\| = |f(Tx)| = |(T^* f)(x)| \leq \|T^* f\| \|x\| \leq \|T^*\| \cdot \|x\|.$$

  The above inequality follows trivially if $Tx = 0$, giving $\|T\| \leq \|T^*\|$ and concluding the claim.

- $(T + \alpha S)^* = T^* + \alpha S^*$ for every $\alpha \in \mathbb{F}$.

- If $T \in \mathcal{B}(V_1, V_2)$ and $S \in \mathcal{B}(V_2, V_3)$, then $(ST)^* = T^* S^*$. In fact,

$$(ST)^* f = f \circ (S \circ T) = (f \circ S) \circ T = T^*(f \circ S) = (T^* \circ S^*) f.$$

- $(id_{V_1})^* = id_{V_1^*}$, so if T is an isomorphism, then T^* is an isomorphism and $(T^*)^{-1} = (T^{-1})^*$.

Example 3.7. Consider the right-shift operator

$$R \colon \ell^1 \to \ell^1, \quad (x_1, x_2, \cdots) \mapsto Rx = (0, x_1, x_2, \cdots).$$

We find R^*.

We know that

$$R^* \colon (\ell^1)^* \to (\ell^1)^*.$$

Take $f \in (\ell^1)^*$, so there exists $b \in \ell^\infty$ such that

$$f(x) = \sum_{n=1}^{\infty} b_n x_n.$$

Then, if $b = (b_1, b_2, \cdots) \in \ell^\infty$ is the representative of the functional f, $L(b) = (b_2, b_3, \cdots)$ is the representative of the functional $R^* f$. Abusing notation, we write, for $b \in \ell^\infty$,

$$R^* b = Lb.$$

3.5 Reflexive Spaces

Definition 3.6. Let V be a normed space. The space $V^* = \mathcal{B}(V, \mathbb{F})$ is the dual of V. The second dual of V is $V^{**} = B(V^*, \mathbb{F})$.

Define the canonical map (the evaluation map)

$$J_V \colon V \to V^{**},\ (J_V x)(f) = f(x),$$

that is,

$$(J_V x, f)_{V^{**},V^*} = (f, x)_{V^*,V}.$$

Proposition 3.2. *J_V is an injective linear isometry, and so $V \cong J(V) \subseteq V^{**}$.*

Proof. Notice that for $x, y \in V$ and $\alpha \in \mathbb{F}$,

$$(J_V(x + \alpha y), f) = f(x + \alpha y) = f(x) + \alpha f(y) = (J_V x, f) + \alpha(J_V y, f) = (J_V(x) + \alpha J_V(y), f),$$

and so J_V is linear.

Moreover, by Hahn–Banach, for $x \neq 0$, $J_V x \neq 0$, and

$$\|J_V x\|_{V^{**}} = \sup_{\substack{f \in V^* \\ \|f\|_* = 1}} |(J_V x, f)| = \sup_{\substack{f \in V^* \\ \|f\|_* = 1}} |f(x)| = \|x\|.$$

We conclude that J_V is an isometry. □

Definition 3.7. We say that V is reflexive if and only if the evaluation map J_V is surjective.

Remark 3.4. We make the following observations.

- All reflexive spaces are isomorphic to their second dual and hence are Banach.
- Reflexivity requires more than V and V^{**} being isometrically isomorphic; it requires that J_V be an isomorphism. In fact, in 1950, James constructed an example of a non-reflexive space that is isometrically isomorphic to its second dual.
- All finite-dimensional spaces are reflexive, since V and V^{**} have the same dimension and J_V is injective and hence surjective.

Example 3.8. ℓ^1 is not reflexive. Indeed, if it were reflexive, then $(\ell^1)^{**} \cong \ell^1$. However, since $(\ell^1)^* \cong \ell^\infty$, we obtain

$$(\ell^\infty)^* \cong \ell^1.$$

Since ℓ^1 is separable, Proposition 3.1 implies that ℓ^∞ is separable, a contradiction.

Example 3.9. For $1 < p < \infty$, ℓ^p is reflexive.

Let

$$J_p \colon \ell^p \to (\ell^p)^{**}.$$

We want to show that J_p is surjective.

For $1 < p < \infty$ and q such that $\dfrac{1}{p} + \dfrac{1}{q} = 1$, we know that the map $\varphi_p : \ell^q \mapsto (\ell^p)^*$, defined by

$$\varphi_p(b) : x \mapsto \sum_{n=1}^{\infty} b_n x_n \qquad b \in \ell^q, x \in \ell^p,$$

is an isomorphism, and so

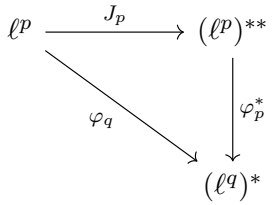
$$\varphi_p^* : (\ell^p)^{**} \mapsto (\ell^q)^*$$

is an isomorphism.

We show that

$$\varphi_p^* \circ J_p = \varphi_q,$$

and conclude that $J_p = (\varphi_p^*)^{-1} \circ \varphi_q$ is surjective.



Let $x \in \ell^p$. We have

$$\begin{aligned}(\varphi_p^*(J_p x), b)_{((\ell^q)^*, \ell^q)} &= (J_p x, \varphi_p(b))_{(\ell^p)^{**}, (\ell^p)^*} \\ &= (\varphi_p(b), x)_{(\ell^p)^*, \ell^p} \\ &= \sum_{n=1}^{\infty} b_n x_n = \sum_{n=1}^{\infty} x_n b_n = (\varphi_q(x), b)_{(\ell^q)^*, \ell^q},\end{aligned}$$

and the claim follows.

Lemma 3.3. *Given two normed spaces V_1, V_2 and $T \in \mathcal{B}(V_1, V_2)$, the following diagram commutes:*

$$\begin{array}{ccc} V_1 & \xrightarrow{T} & V_2 \\ {\scriptstyle J_{V_1}}\downarrow & & \downarrow{\scriptstyle J_{V_2}} \\ V_1^{**} & \xrightarrow[T^{**}]{} & V_2^{**} \end{array}$$

That is,

$$T^{**} \circ J_{V_1} = J_{V_2} \circ T.$$

Proof.

$$(T^{**} \circ J_{V_1} x, g)_{V_2^{**}, V_2^*} = (J_{V_1} x, T^* g)_{V_1^{**}, V_1^*} = (T^* g, x)_{V_1^*, V_1} = (g, Tx)_{V_2^*, V_2} = (J_{V_2}(Tx), g)_{V_2^{**}, V_2^*}.$$

□

Corollary 3.6. *If $V_1 \cong V_2$, then V_1 is reflexive $\iff$ V_2 is reflexive.*

Proof. Let $T : V_1 \mapsto V_2$ be an isometric isomorphism. Then T^{-1} is also an isometric isomorphism. We also obtain that T^{**} is an isometric isomorphism, as is $(T^{**})^{-1}$.

From the previous lemma, we have

$$J_{V_1} = (T^{**})^{-1} \circ J_{V_2} \circ T, \qquad J_{V_2} = T^{**} \circ J_{V_1} \circ T^{-1},$$

which gives the desired equivalence. □

Lemma 3.4. *A closed subspace of a reflexive space is reflexive.*

Proof. Let $\mathcal{M}$ be a closed subspace of a reflexive space V. Without loss of generality, assume that $\mathcal{M}$ is a proper subspace.

Define the canonical embeddings

$$J_V : V \mapsto V^{**} \qquad\qquad J_{\mathcal{M}} : \mathcal{M} \mapsto \mathcal{M}^{**}.$$

We know that J_V is surjective, and we want to show that $J_{\mathcal{M}}$ is surjective.

Let $\xi \in \mathcal{M}^{**}$, and define the linear functional $\eta : V^* \mapsto \mathbb{F}$ by

$$\eta(f) = \xi(f_{|\mathcal{M}}).$$

Notice that

$$|\eta(f)| \leq \|\xi\|_{**}\|f_{|\mathcal{M}}\|_* \leq \|\xi\|_{**}\|f\|_*.$$

Hence, η is bounded, that is, $\eta \in V^{**}$. Therefore, there exists $x_0 \in V$ such that $J_V x_0 = \eta$.

We show that $x_0 \in \mathcal{M}$. Suppose not. Then, by Hahn–Banach, there exists $f \in V^*$ such that $f_{|\mathcal{M}} = 0$ and $f(x_0) = 1$. Hence,

$$(J_V(x_0), f) = f(x_0) = 1.$$

On the other hand,

$$(J_V(x_0), f) = (\eta, f) = (\xi, f|_{\mathcal{M}}) = \xi(0) = 0,$$

a contradiction.

It remains to show that $J_{\mathcal{M}} x_0 = \xi$. Let $f \in \mathcal{M}^*$, and let $F \in V^*$ be a Hahn–Banach extension of f. Then

$$(J_{\mathcal{M}} x_0, f) = f(x_0) = F(x_0) = (J_V x_0, F) = (\eta, F) = (\xi, F_{|\mathcal{M}}) = (\xi, f).$$

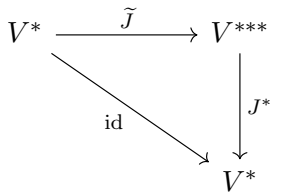
□

We conclude this chapter with the following characterization of reflexive spaces.

Lemma 3.5. *Let V be a normed vector space, and let $J_V : V \to V^{**}$ and $\widetilde{J}_{V^*} : V^* \to V^{***}$ be the canonical embeddings. Then*

$$(J_V)^* \circ J_{V^*} = \mathrm{id}_{V^*},$$

where $(J_V)^* : V^{***} \to V^*$ is the adjoint of J. Equivalently, the following diagram commutes:



Proof. For every $f \in V^*$ and $x \in V$,

$$((J_V^* \circ J_{V^*})f, x)_{V^*,V} = (J_{V^*}(f), Jx)_{V^{***},V^{**}} = (Jx, f)_{V^{**},V^*} = (f, x)_{V^*,V}.$$

□

Remark 3.5. Notice that the above lemma implies that the adjoint map $J_V^* : V^{***} \mapsto V^*$ is always surjective. Be careful: this is not true in general. That is, if a map T is injective, it does not necessarily follow that T^* is surjective. We will revisit this later.

Theorem 3.4. *Given a normed space V, the space V is reflexive if and only if V is Banach and V^* is reflexive.*

Remark 3.6. As a consequence of the above theorem, since ℓ^1 is Banach but not reflexive, we deduce that $(\ell^1)^*$ is not reflexive. However, $(\ell^1)^* \cong \ell^\infty$, and so ℓ^∞ is not reflexive. This completes our analysis of the reflexivity of the ℓ^p spaces.

Proof of ($\implies$). Since V is reflexive, it is Banach. To show that V^* is reflexive, define the canonical embeddings

$$J_V : V \mapsto V^{**} \qquad\qquad J_{V^*} : V^* \mapsto V^{***}.$$

Since J_V is an isometric isomorphism, J_V^* is also an isometric isomorphism. Hence, by the previous lemma, $J_{V^*} = (J_V^*)^{-1}$ is an isometric isomorphism. □

Proof of ($\impliedby$). We have $V \cong J(V)$, so it is enough to show that $J(V)$ is reflexive. We showed in the previous implication that the dual of a reflexive space is reflexive. Since V^* is reflexive, it follows that V^{**} is reflexive.

Hence, it is enough to show that $J(V)$ is a closed subspace of V^{**}. But J is an isometry so done.

□